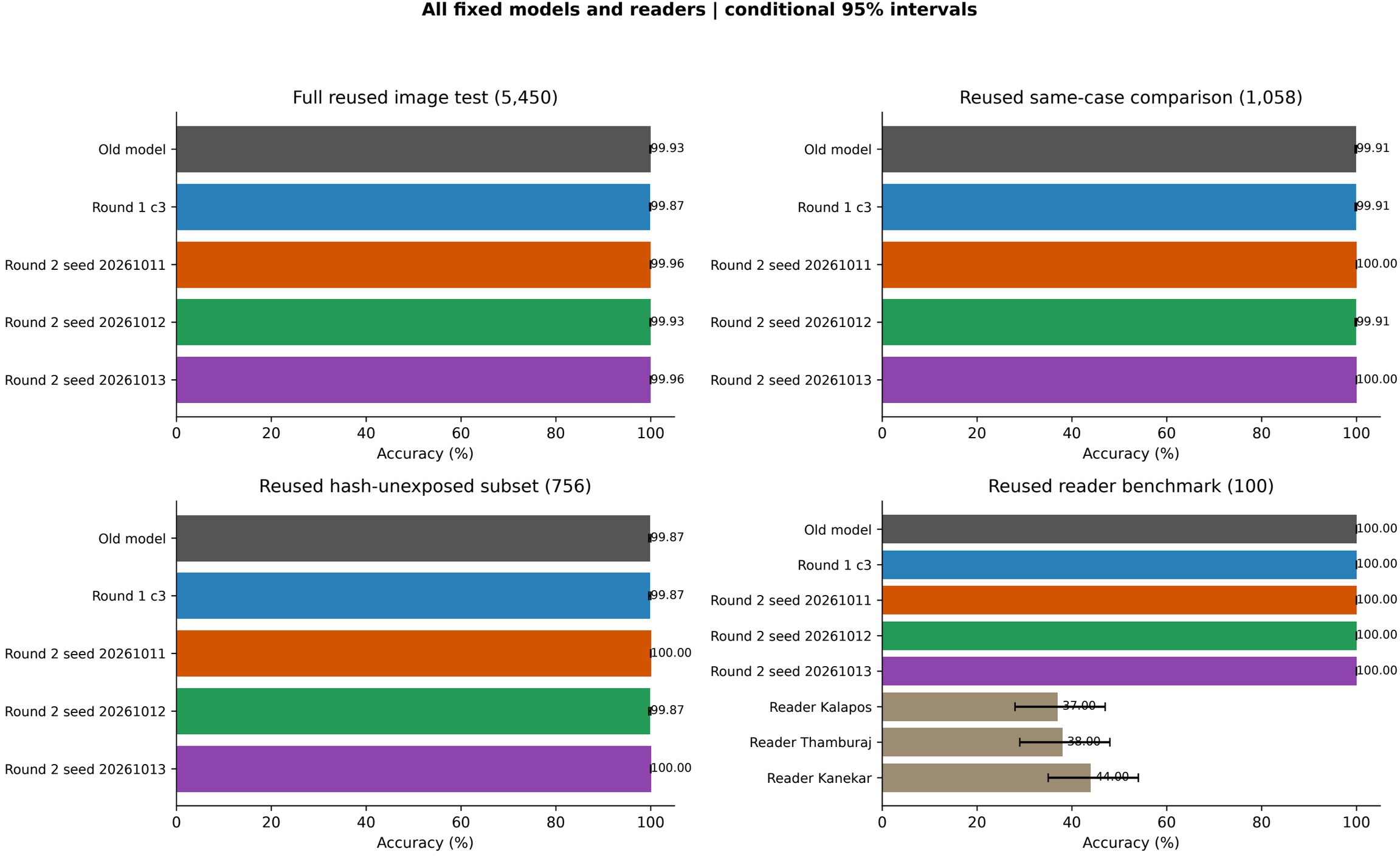
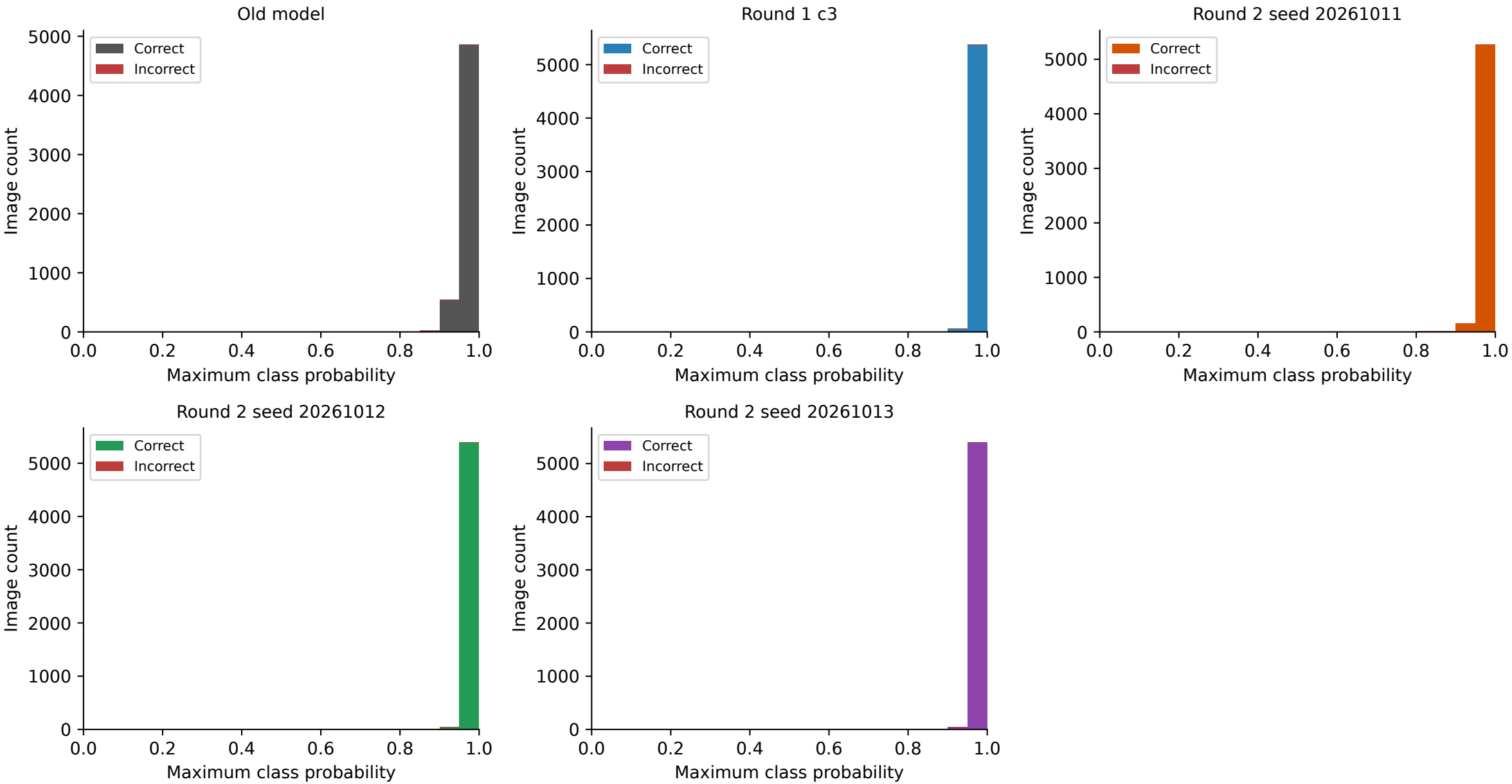


Paired 2,000-draw hash-group bootstrap; original100 image units. Degenerate intervals do not prove certainty.
Three-reader range is descriptive across fixed readers; repeated benchmarks cannot establish independent superiority.

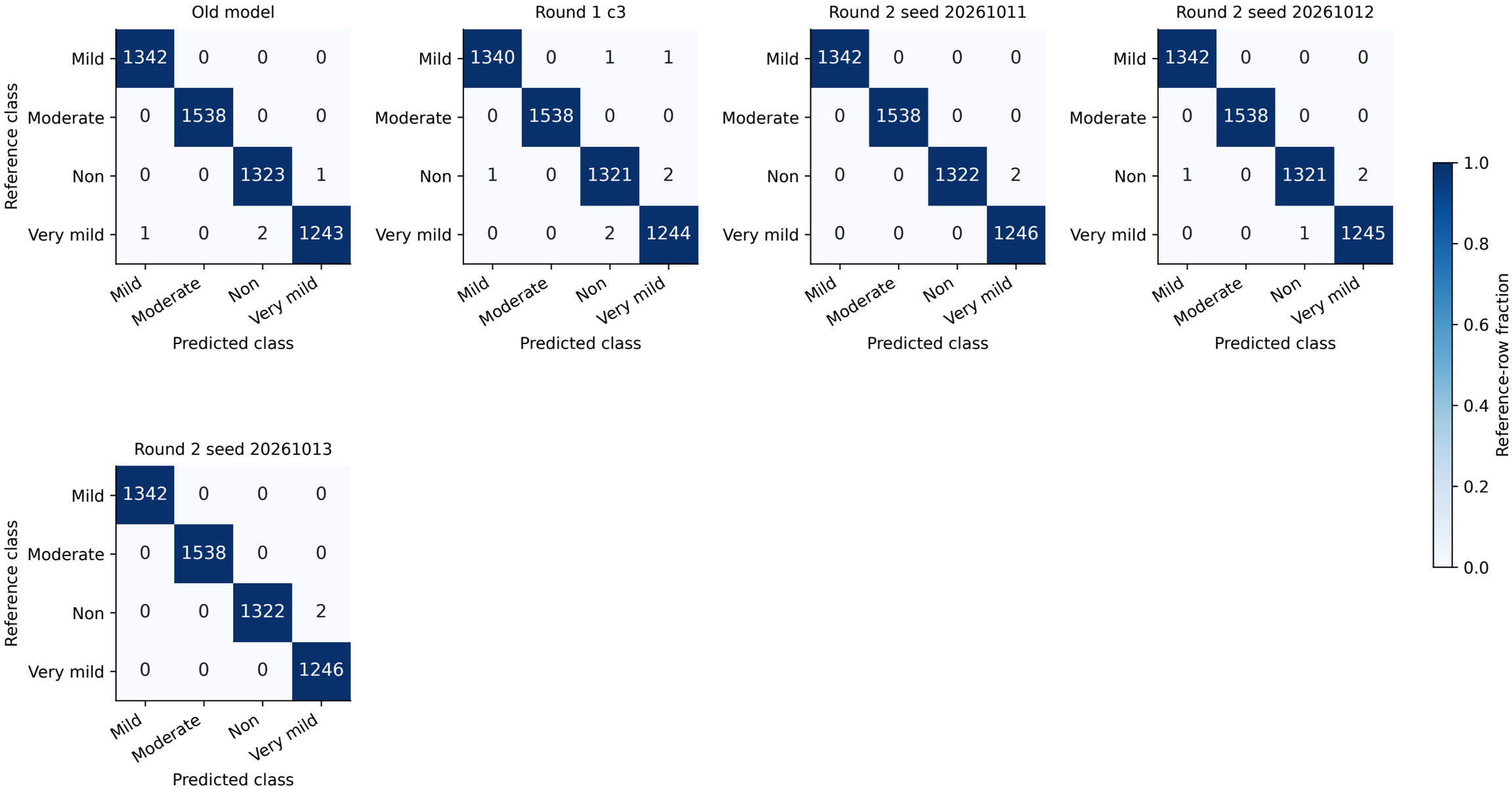
Full reused image test (5,450) | Confidence



Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

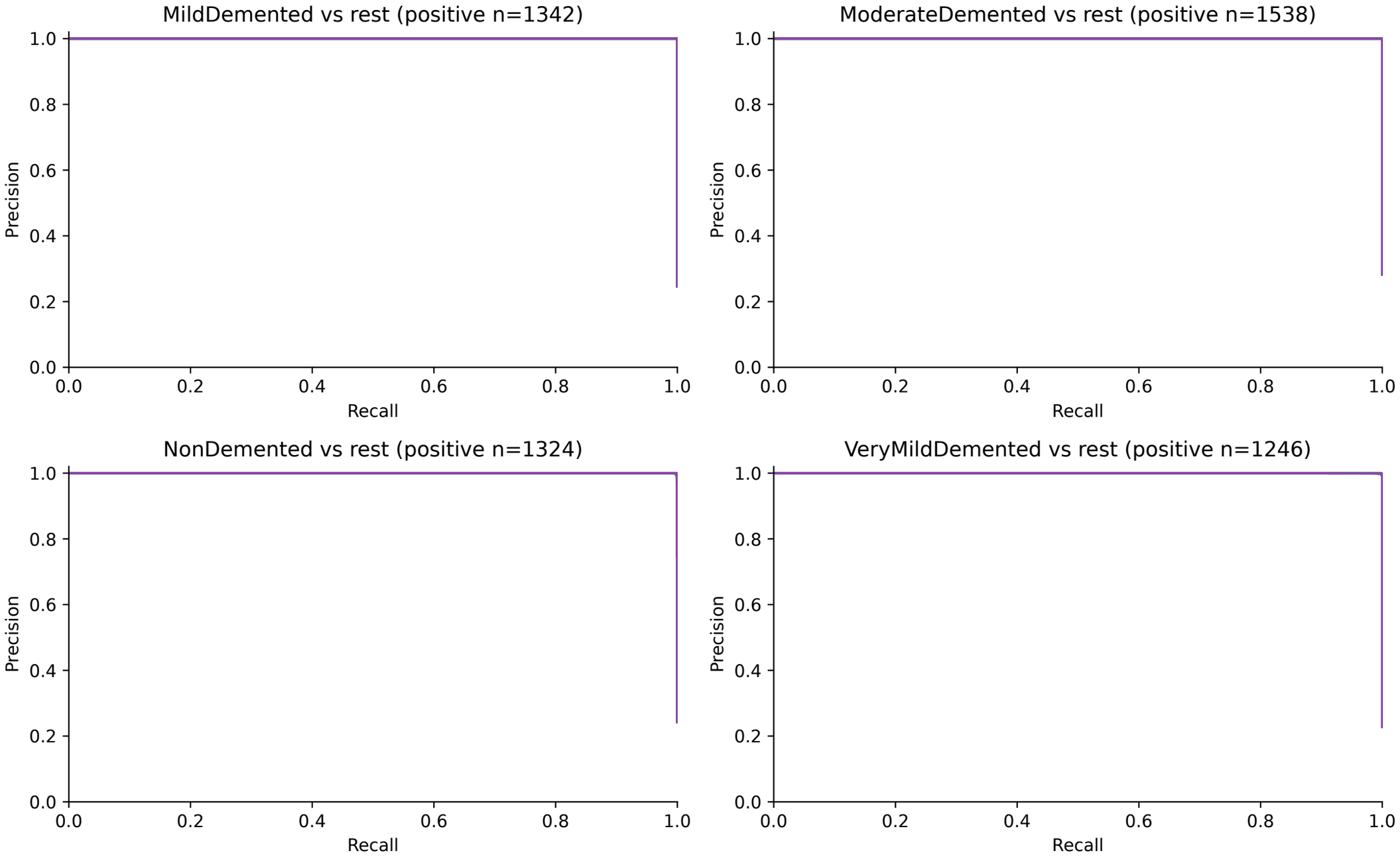
Full reused image test (5,450) | confusion | n=5450

Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.

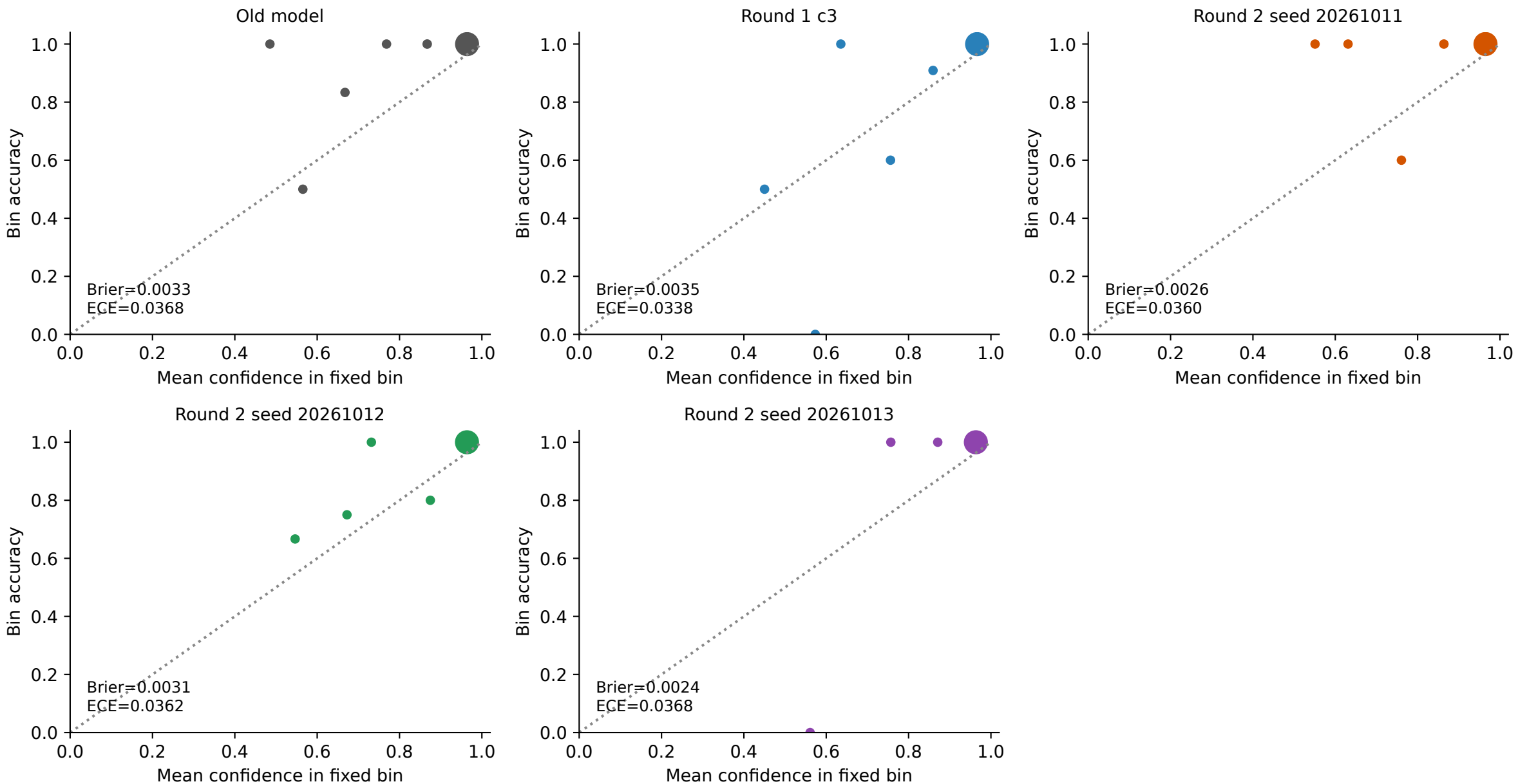


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

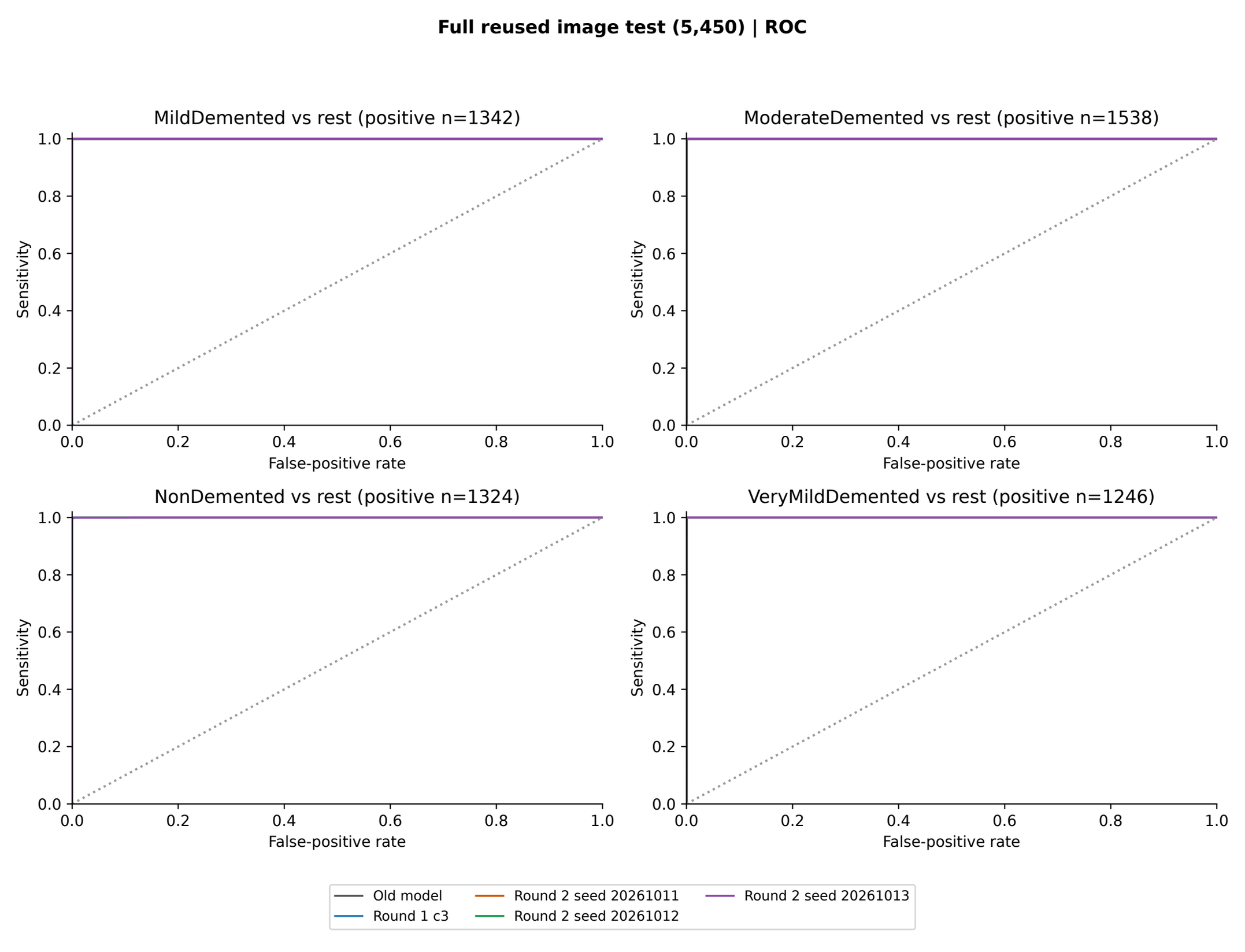
Full reused image test (5,450) | Precision–recall

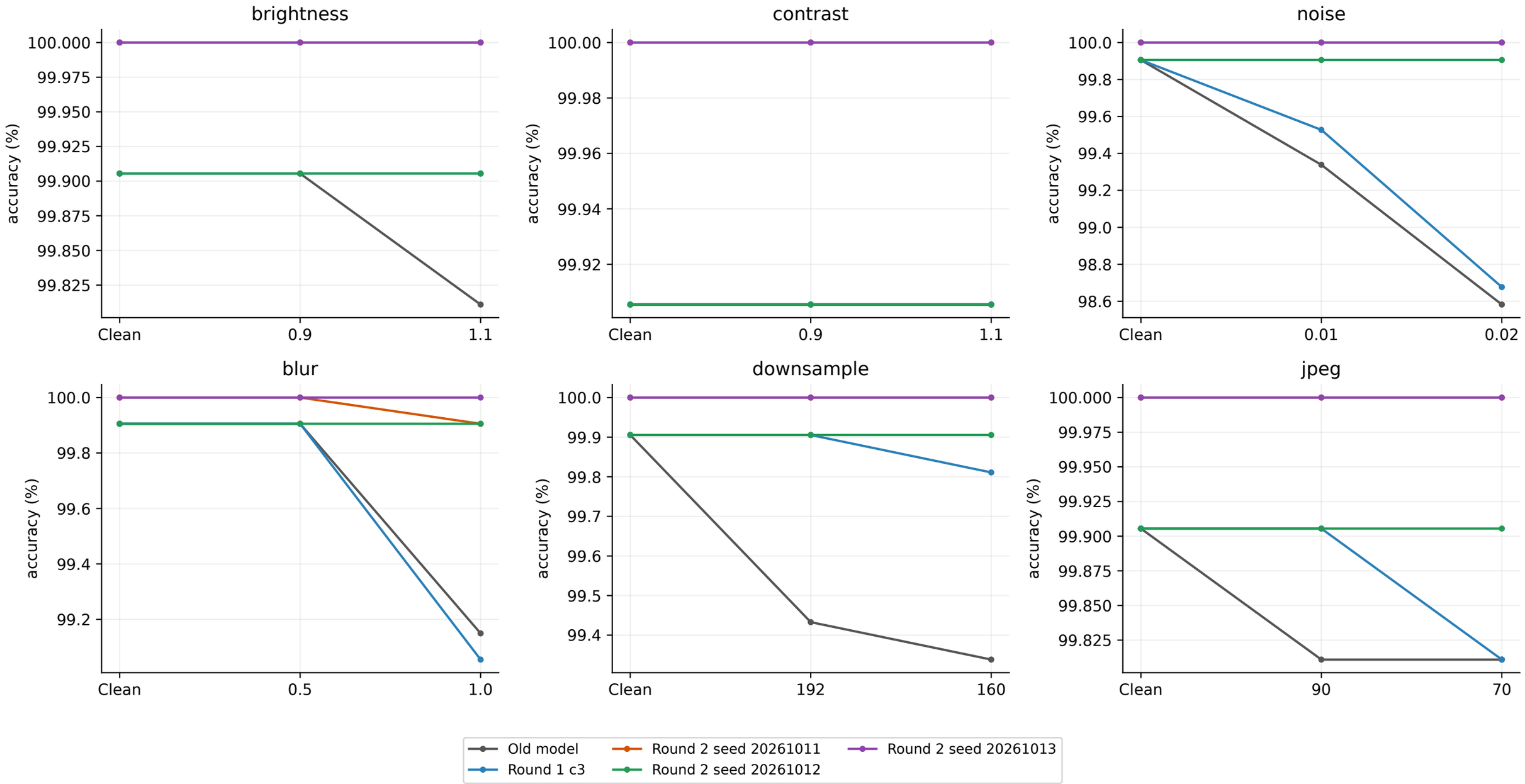


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

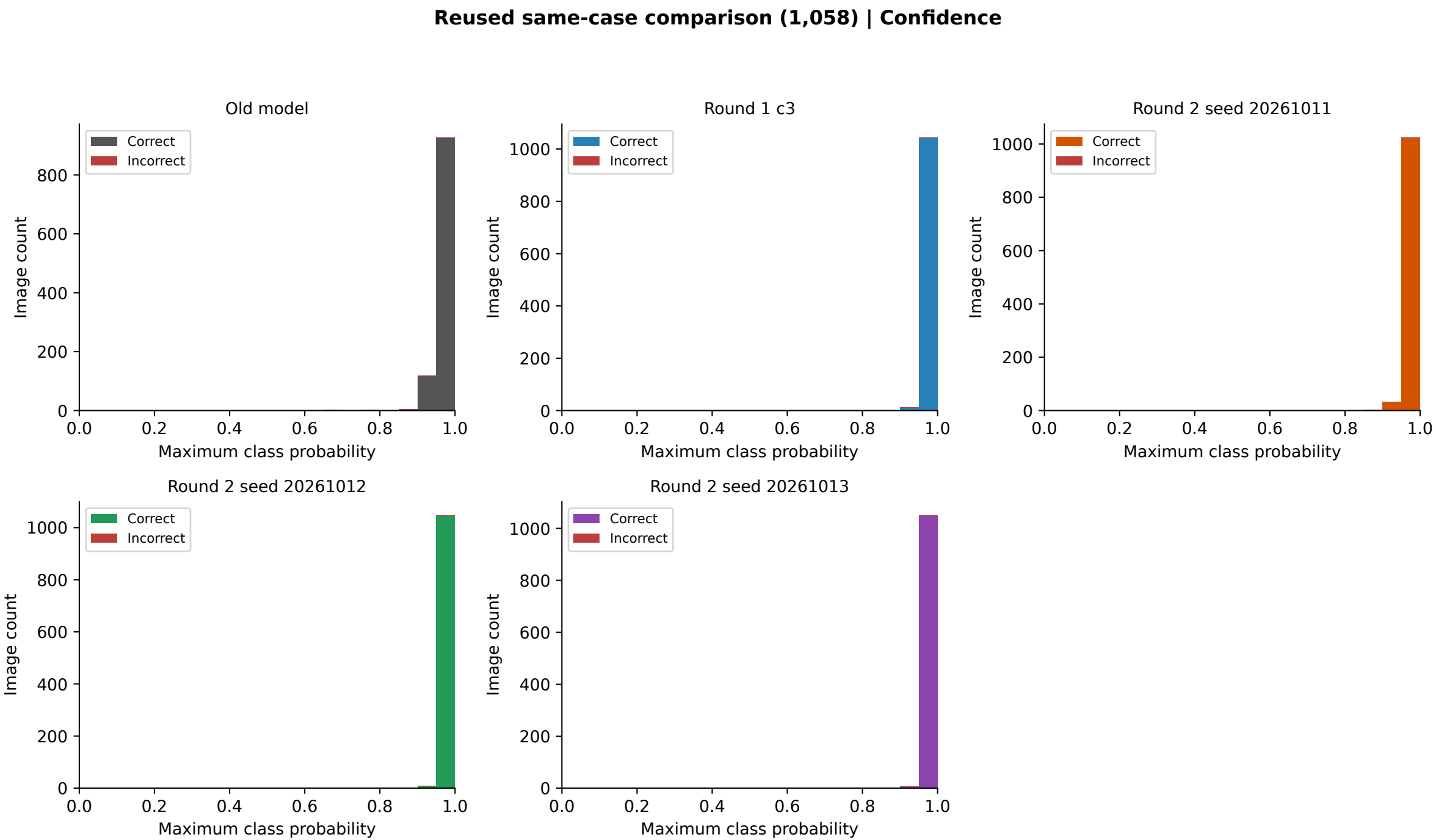


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Full reused image test (5,450) | ROC
MildDemented vs rest (positive n=1342)
ModerateDemented vs rest (positive n=1538)
NonDemented vs rest (positive n=1324)
VeryMildDemented vs rest (positive n=1246)
Sensitivity
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.



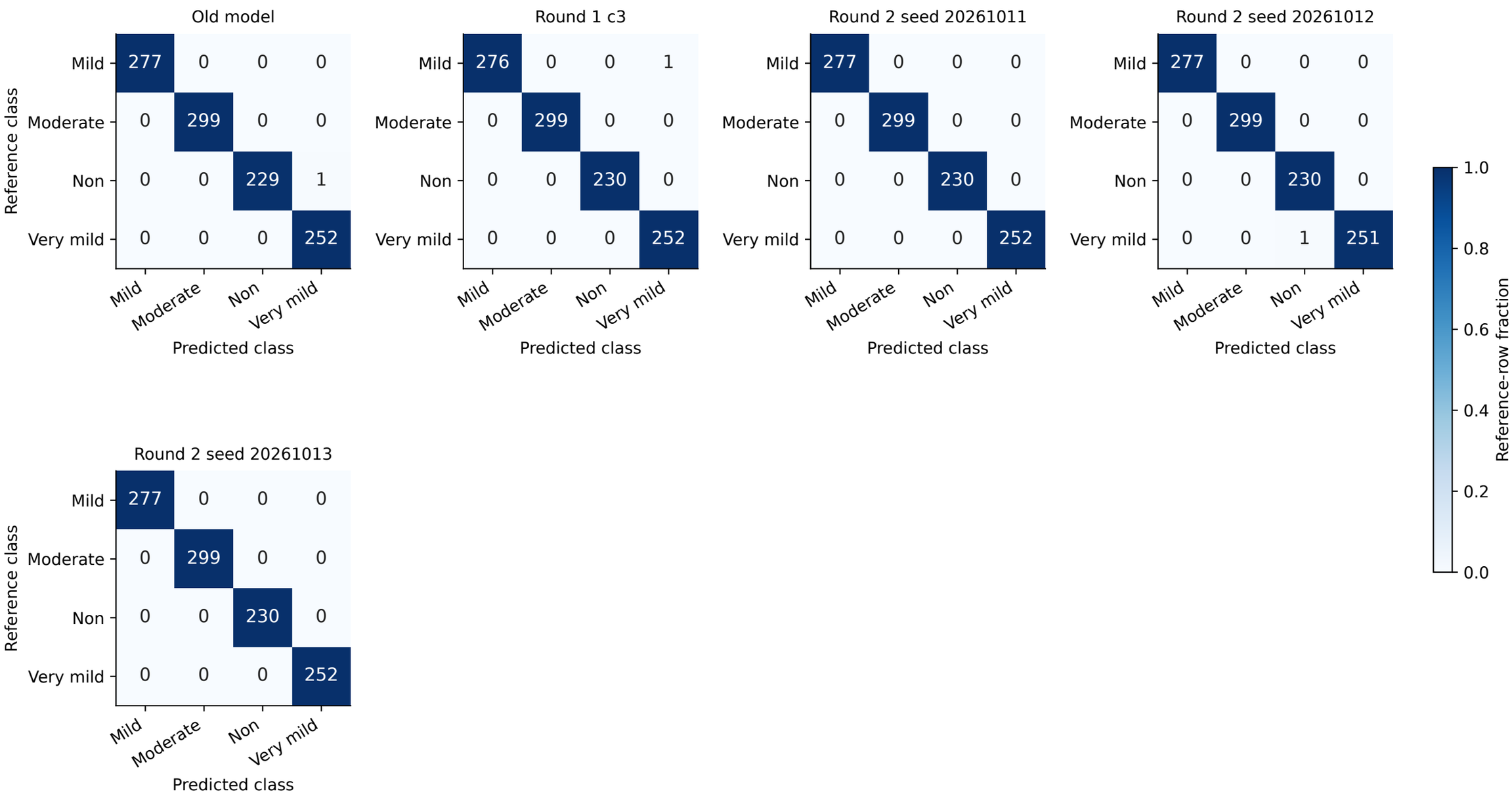
Same-device clean reference; all13 unchanged scenarios. Synthetic pixel stress is not MRI acquisition or clinical validation.



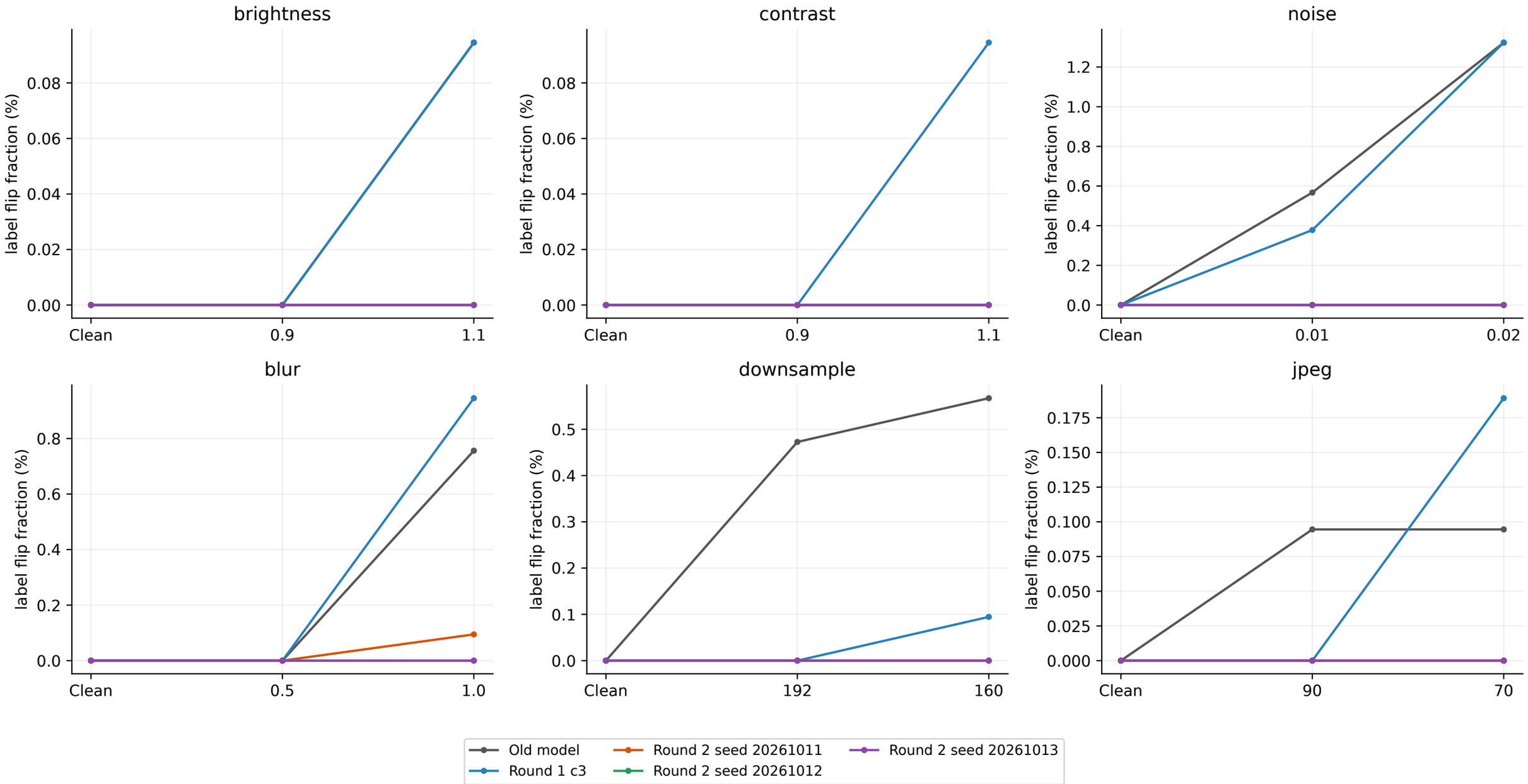
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

Reused same-case comparison (1,058) | confusion | n=1058

Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.

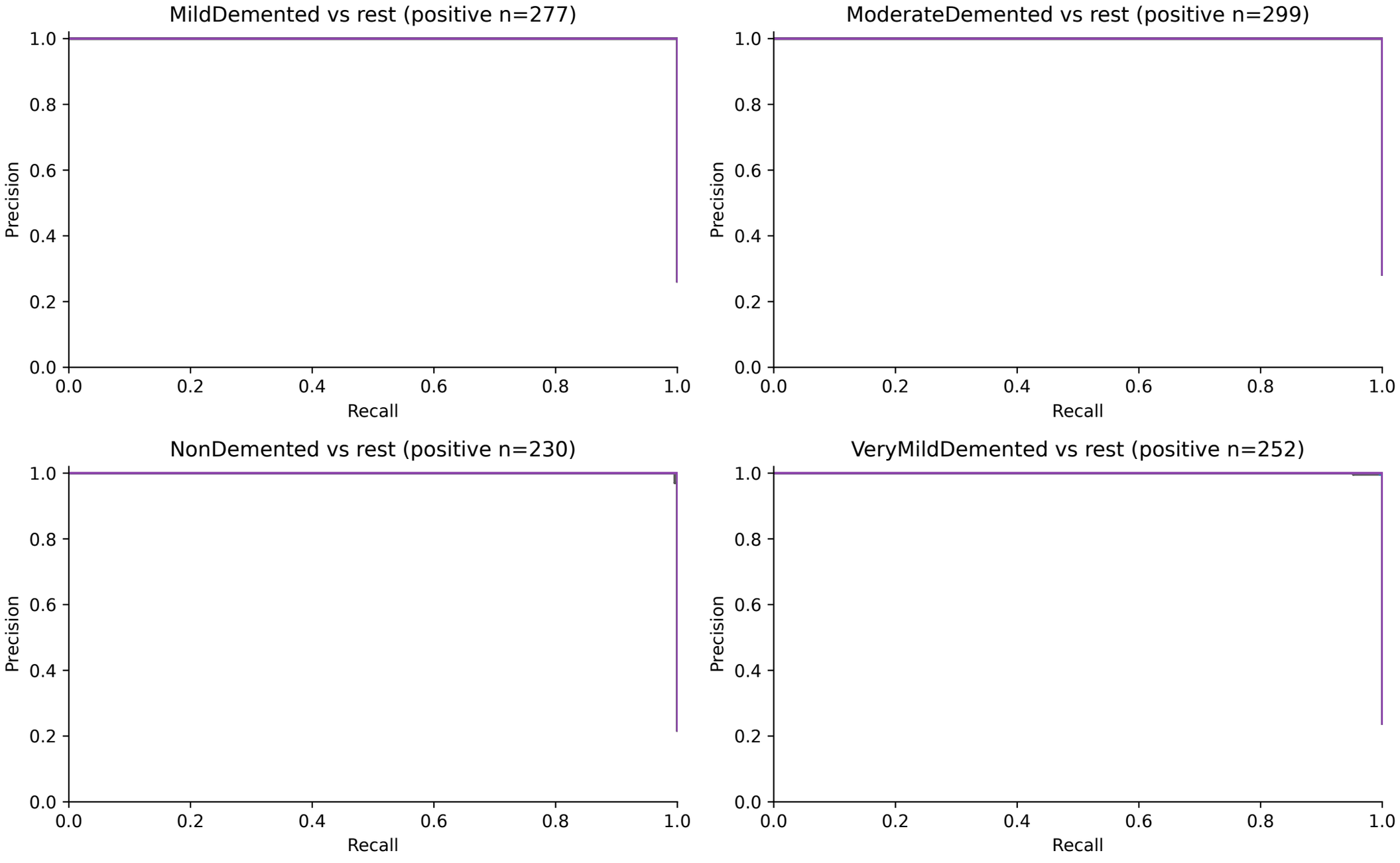


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

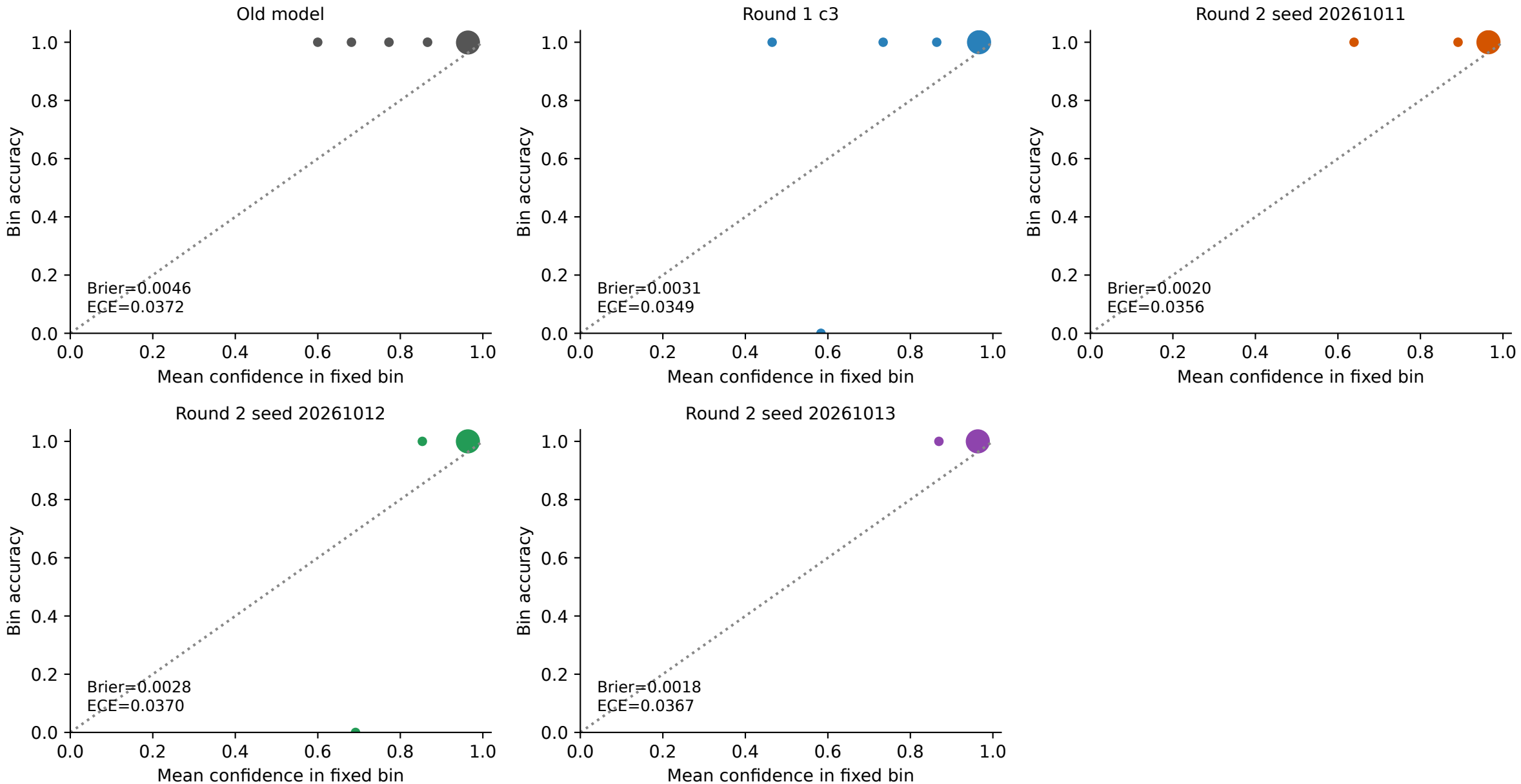


Same-device clean reference; all13 unchanged scenarios. Synthetic pixel stress is not MRI acquisition or clinical validation.

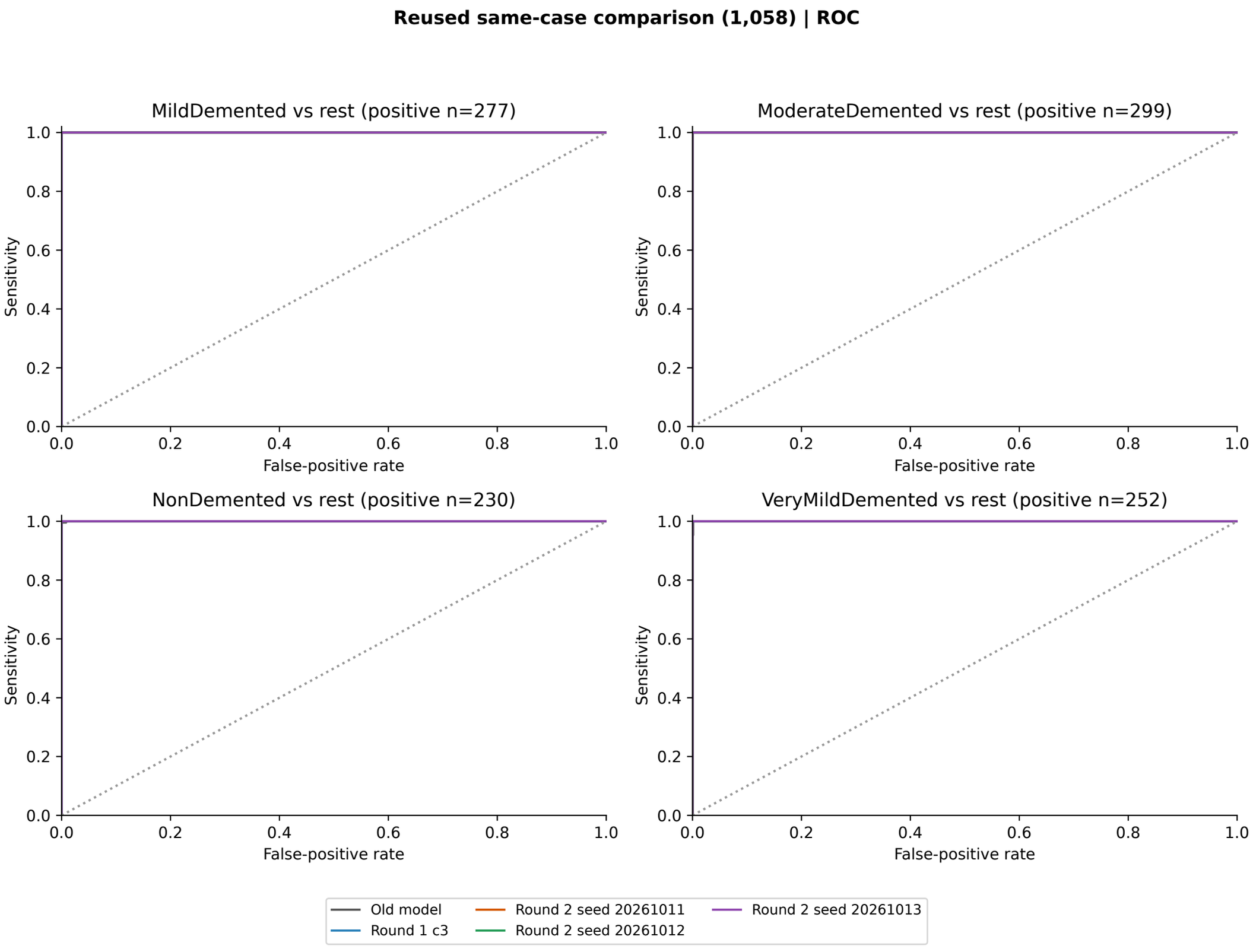
Reused same-case comparison (1,058) | Precision–recall



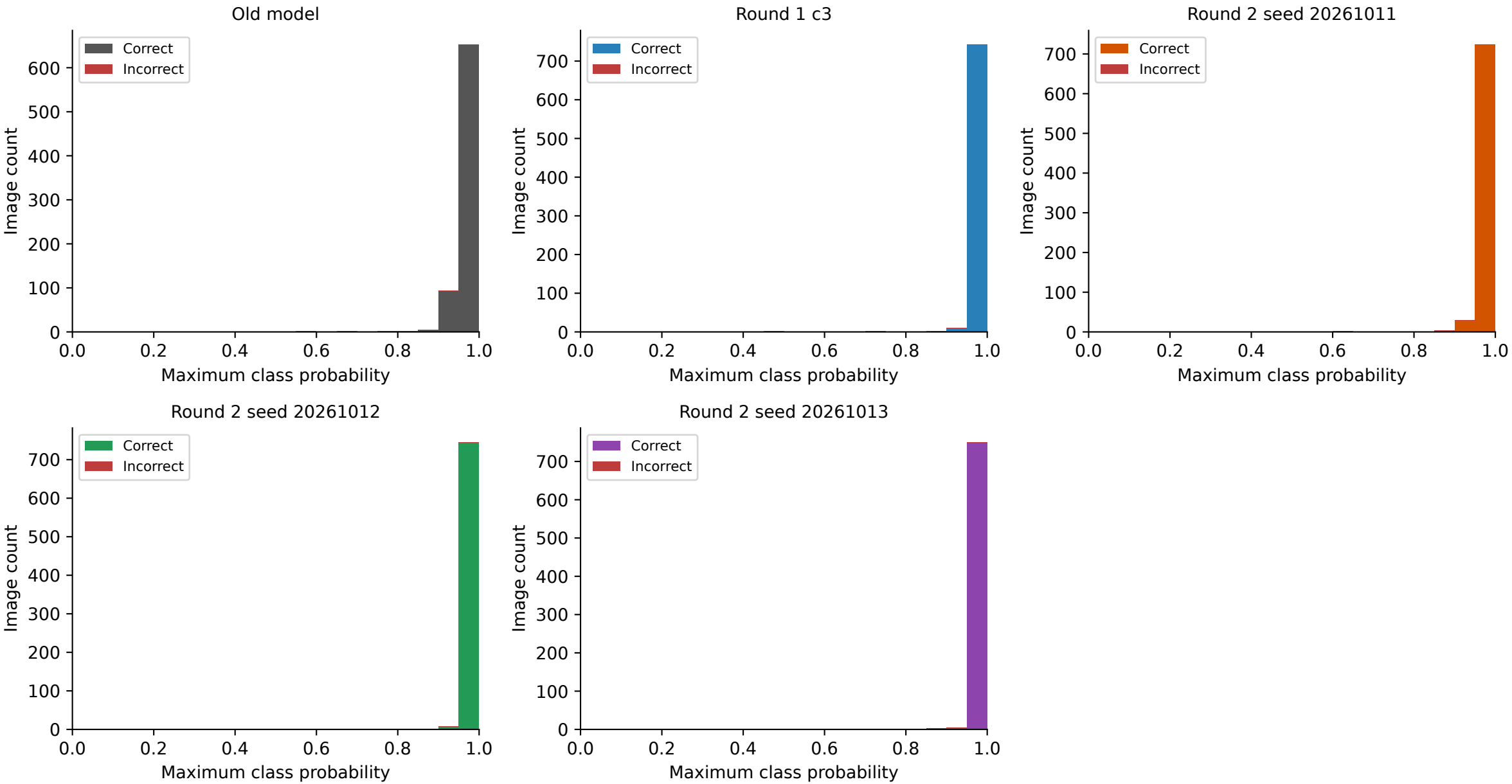
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.



Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Reused same-case comparison (1,058) | ROC
MildDemented vs rest (positive n=277)
ModerateDemented vs rest (positive n=299)
NonDemented vs rest (positive n=230)
VeryMildDemented vs rest (positive n=252)
Sensitivity
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

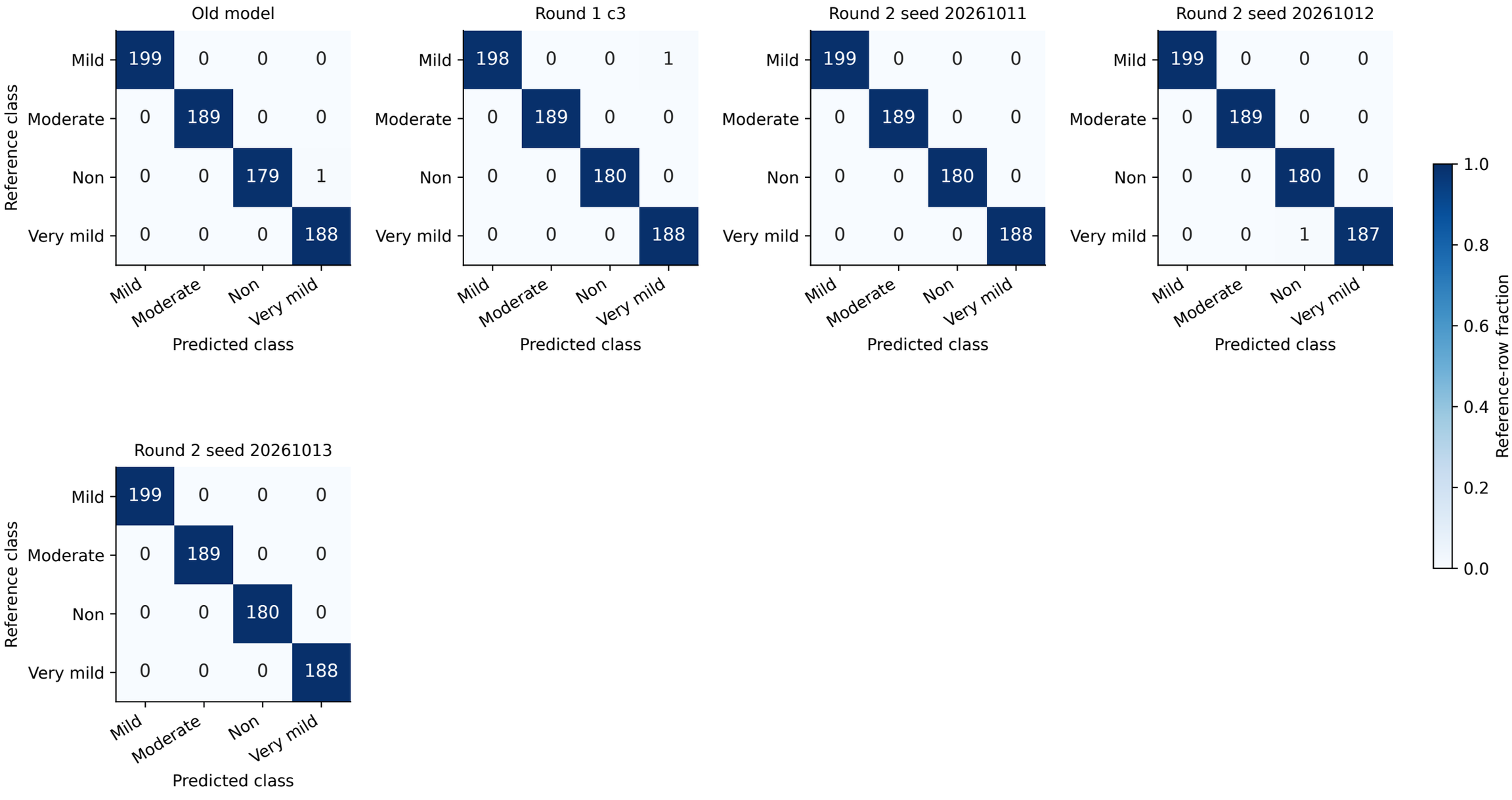
Reused hash-unexposed subset (756) | Confidence



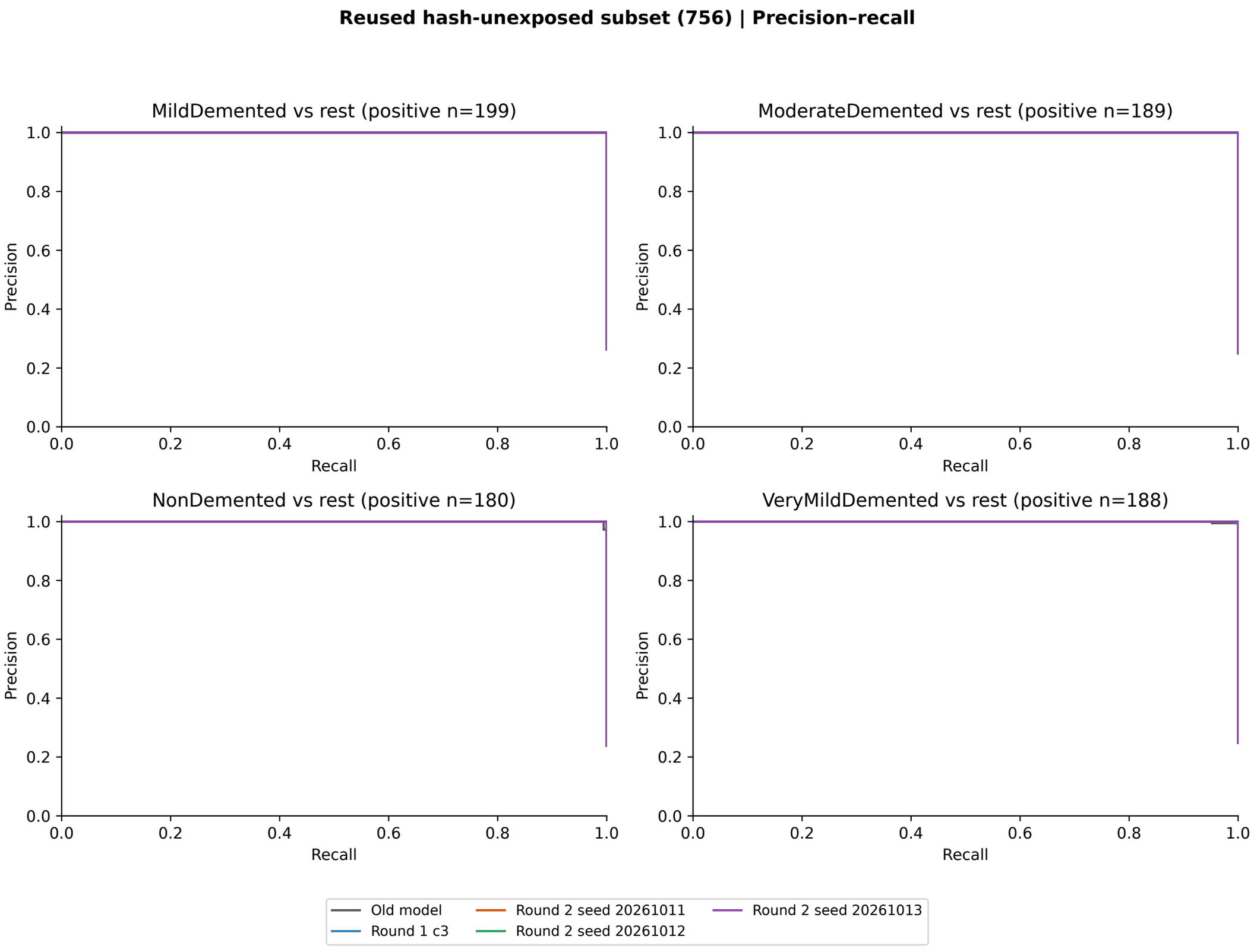
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

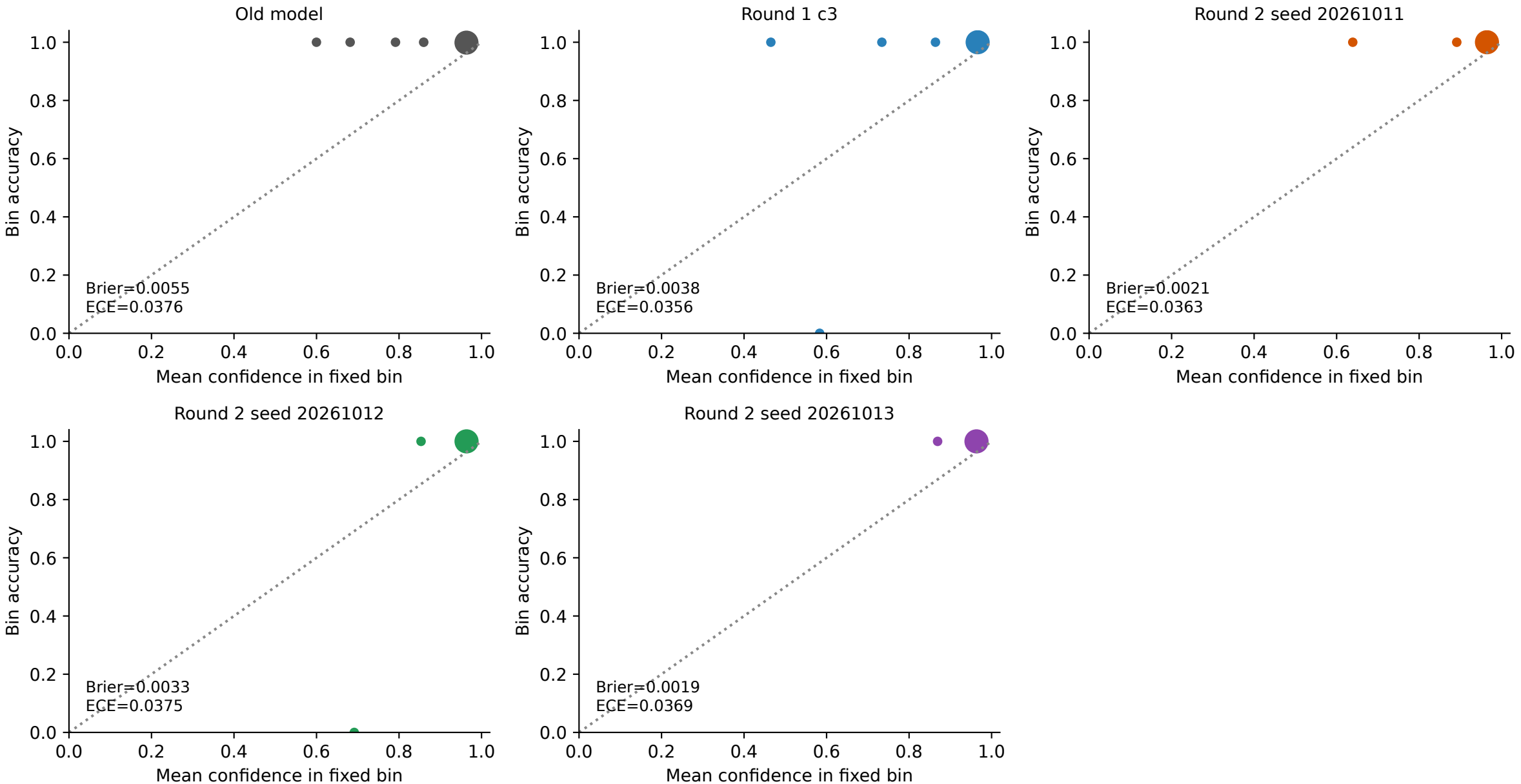
Reused hash-unexposed subset (756) | confusion | n=756

Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.

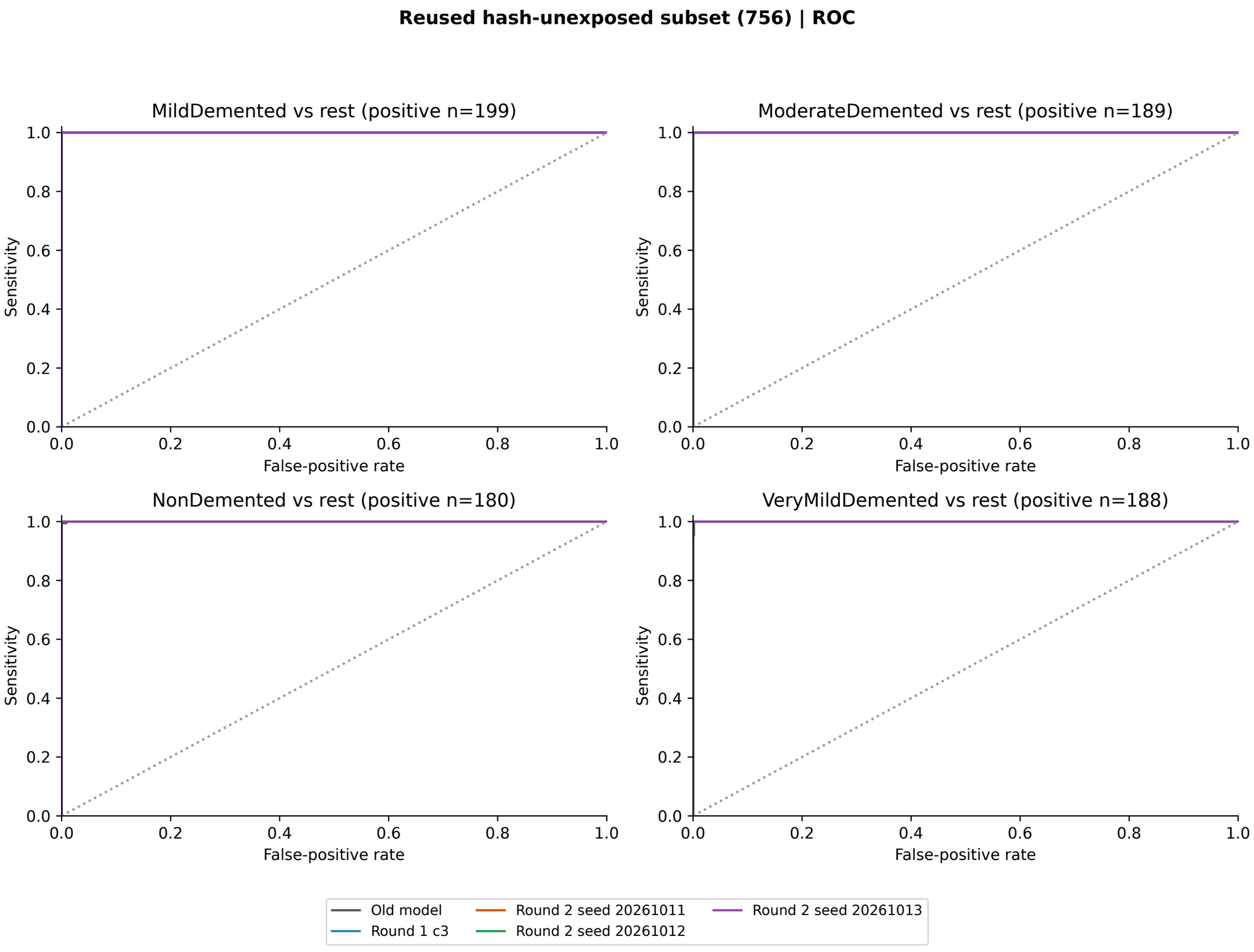


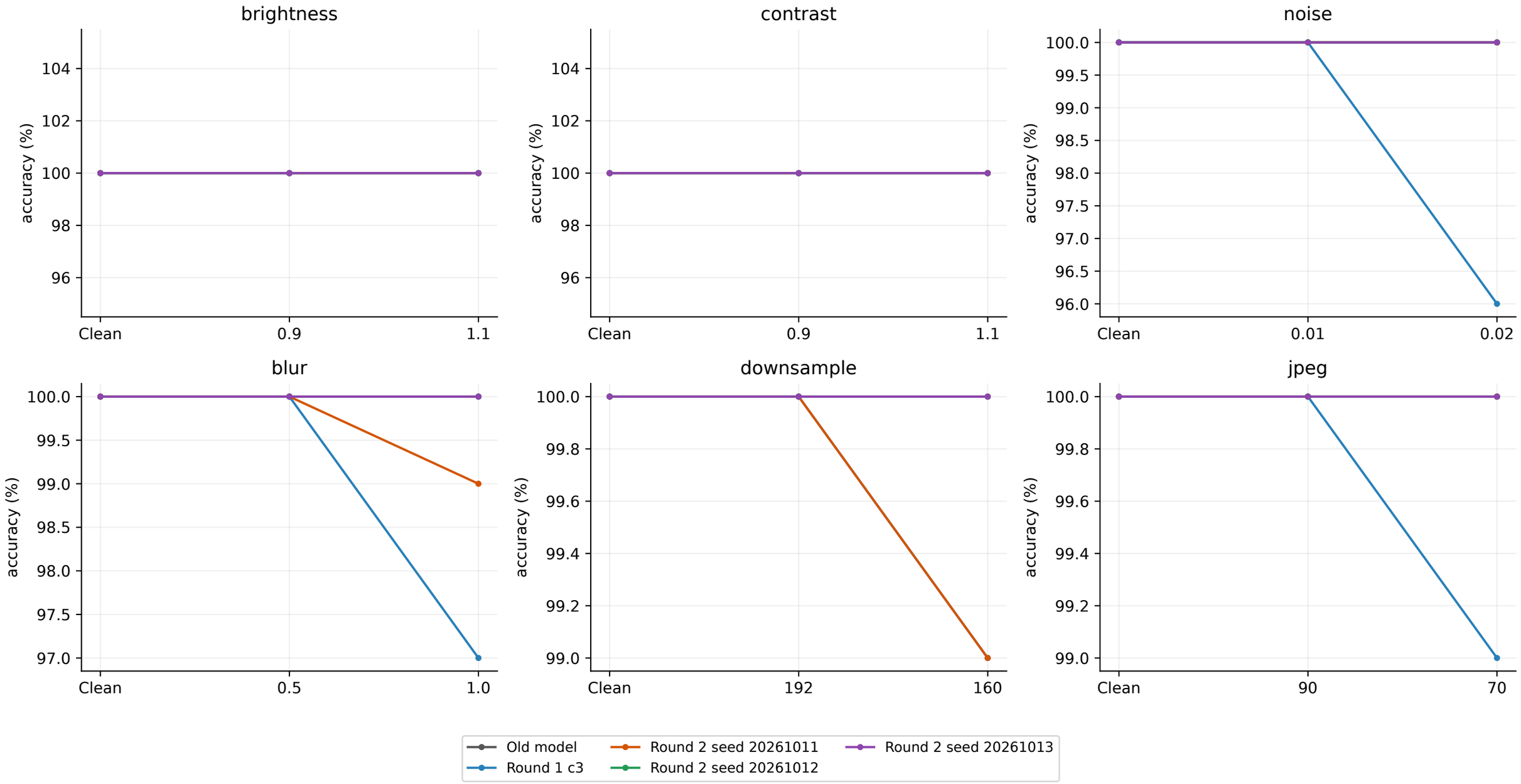
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Reused hash-unexposed subset (756) | Precision–recall
MildDemented vs rest (positive n=199)
ModerateDemented vs rest (positive n=189)
NonDemented vs rest (positive n=180)
VeryMildDemented vs rest (positive n=188)
Precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.



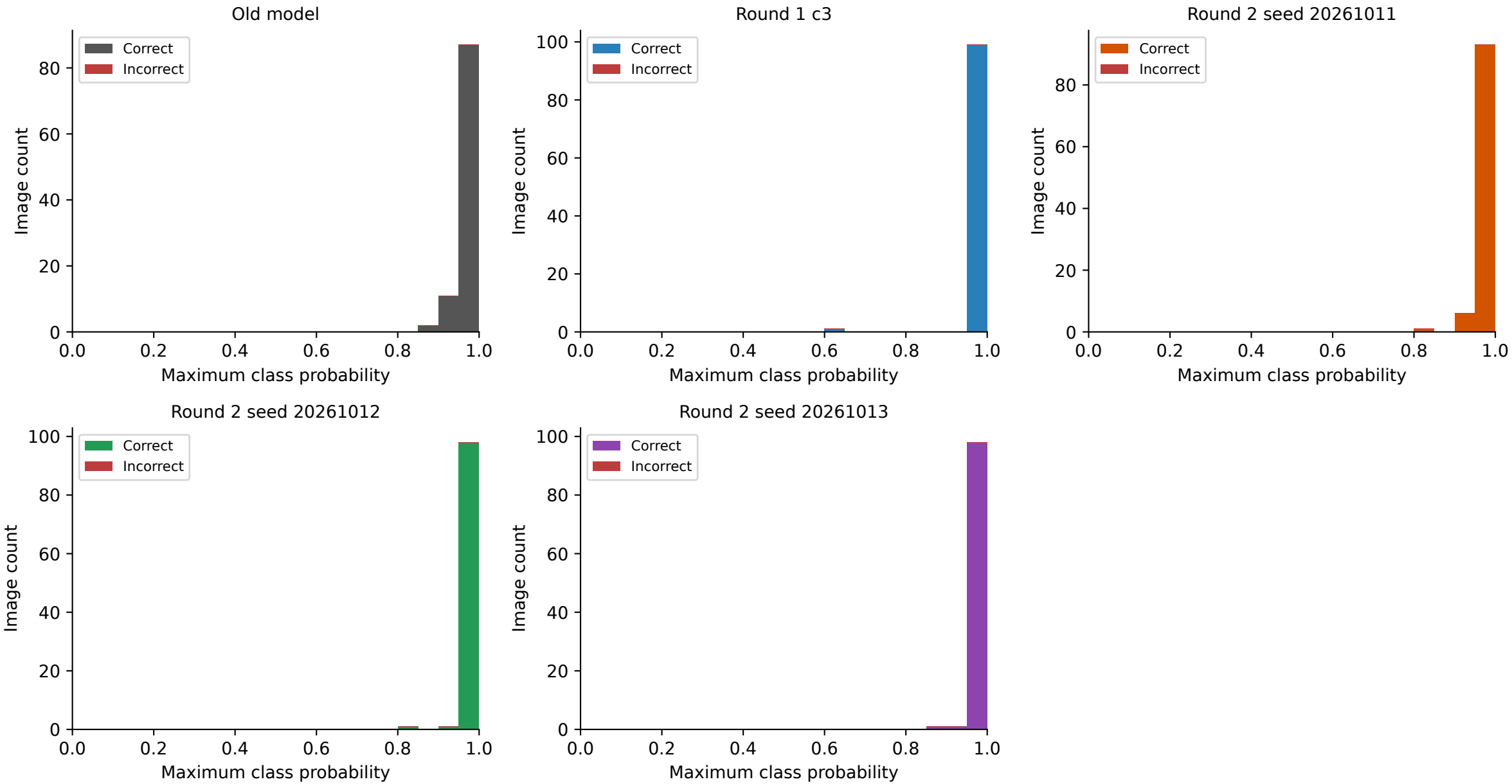
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Reused hash-unexposed subset (756) | ROC
MildDemented vs rest (positive n=199)
ModerateDemented vs rest (positive n=189)
NonDemented vs rest (positive n=180)
VeryMildDemented vs rest (positive n=188)
Sensitivity
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

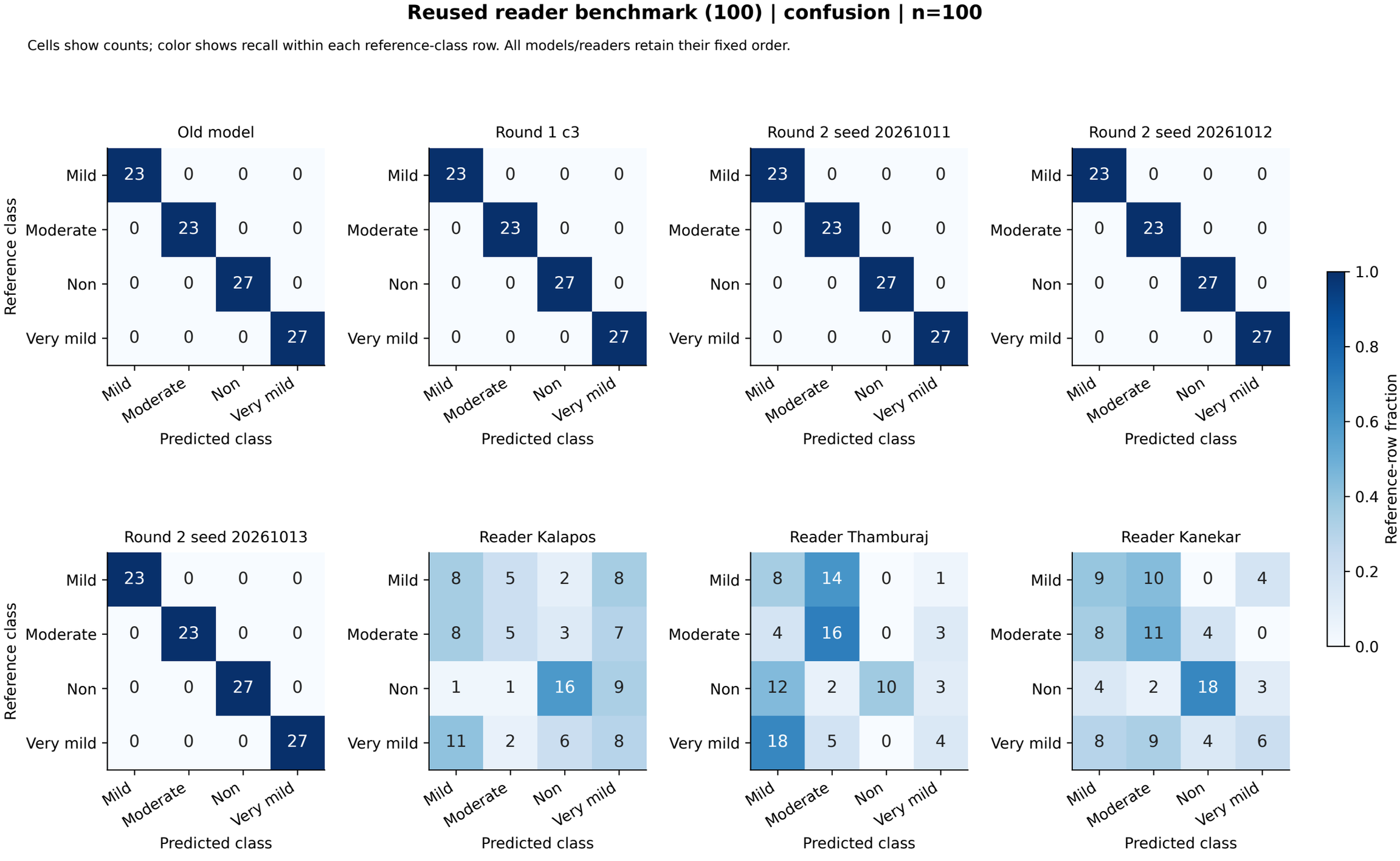


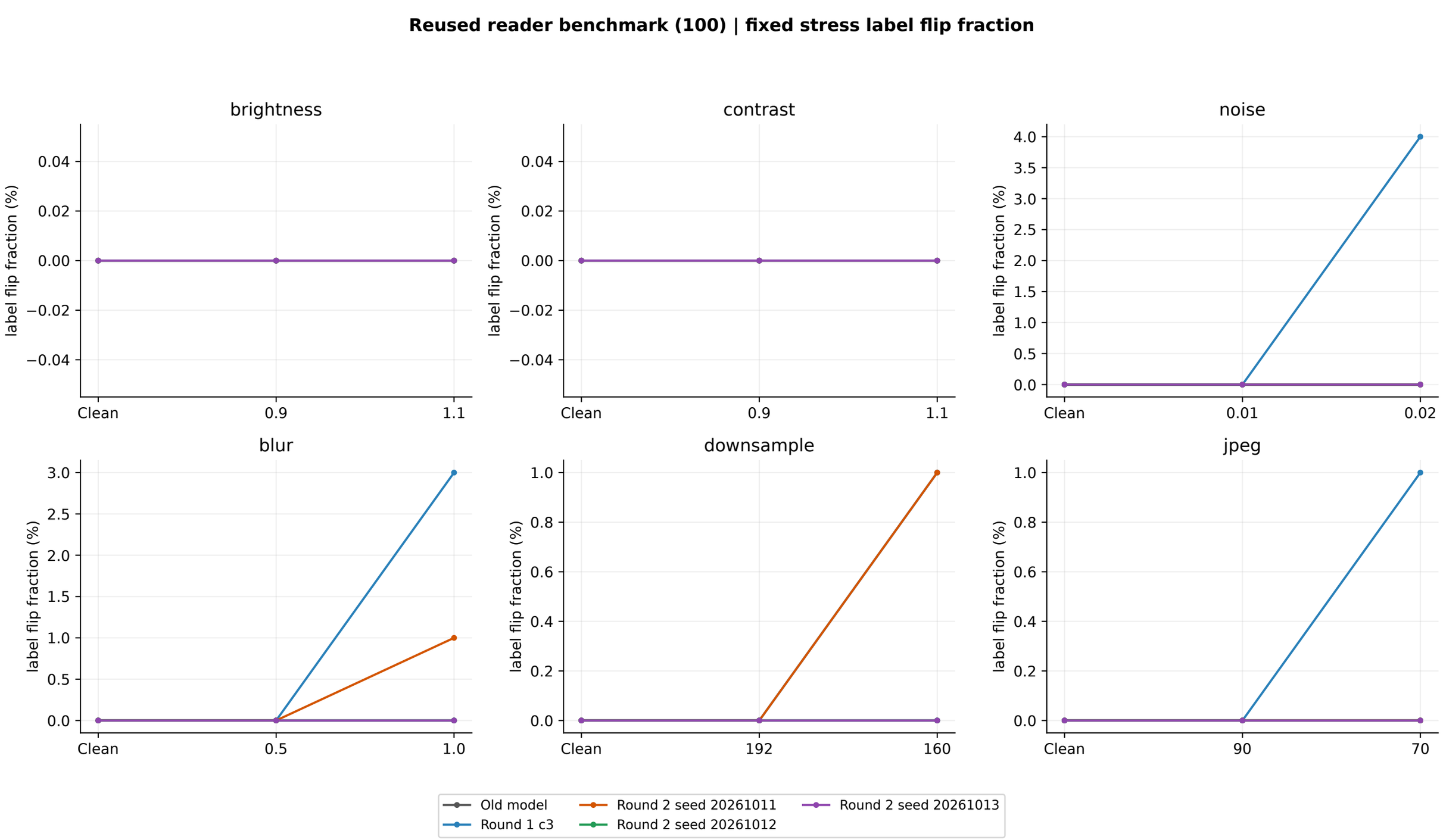
Same-device clean reference; all13 unchanged scenarios. Synthetic pixel stress is not MRI acquisition or clinical validation.

Reused reader benchmark (100) | Confidence

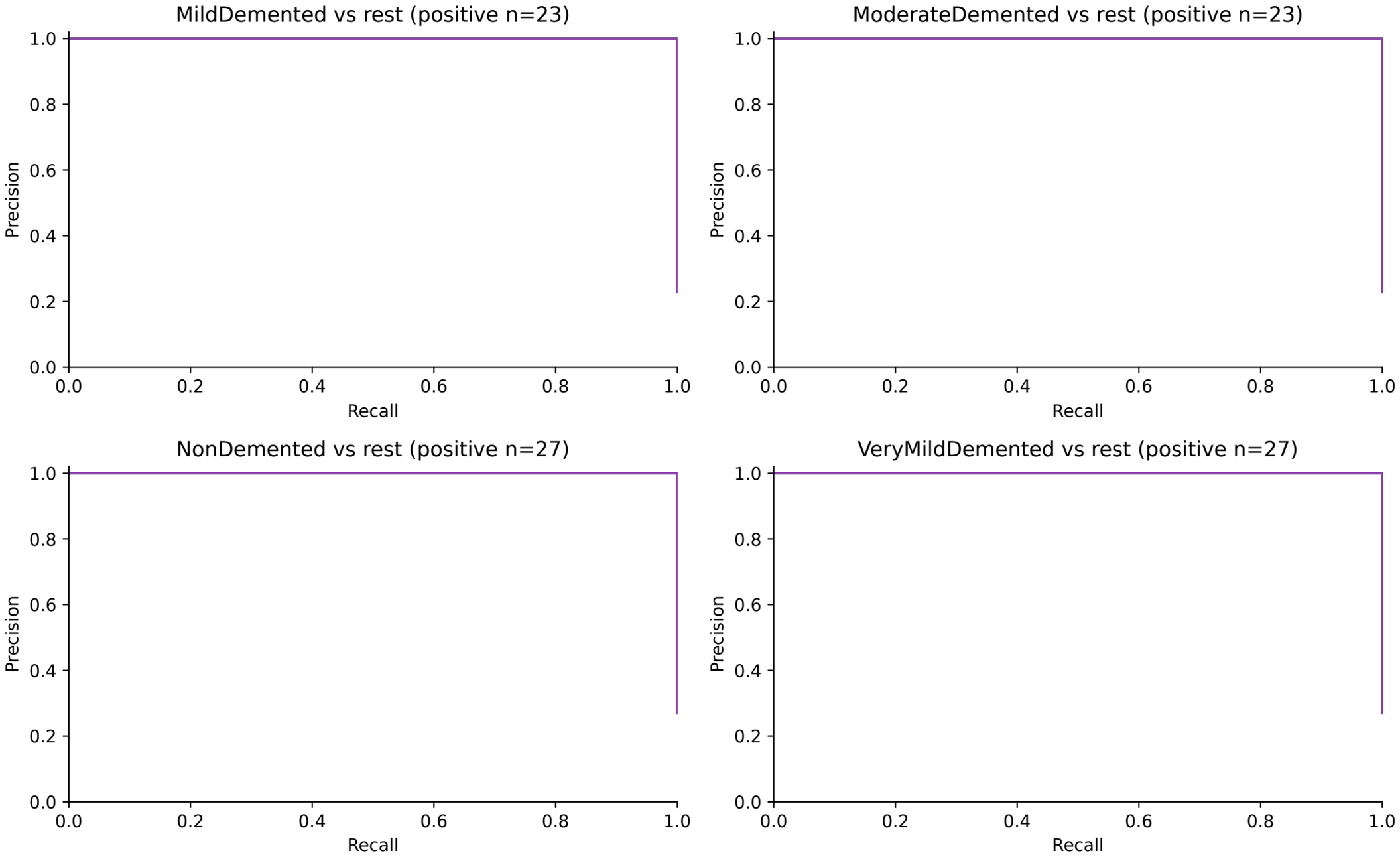


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

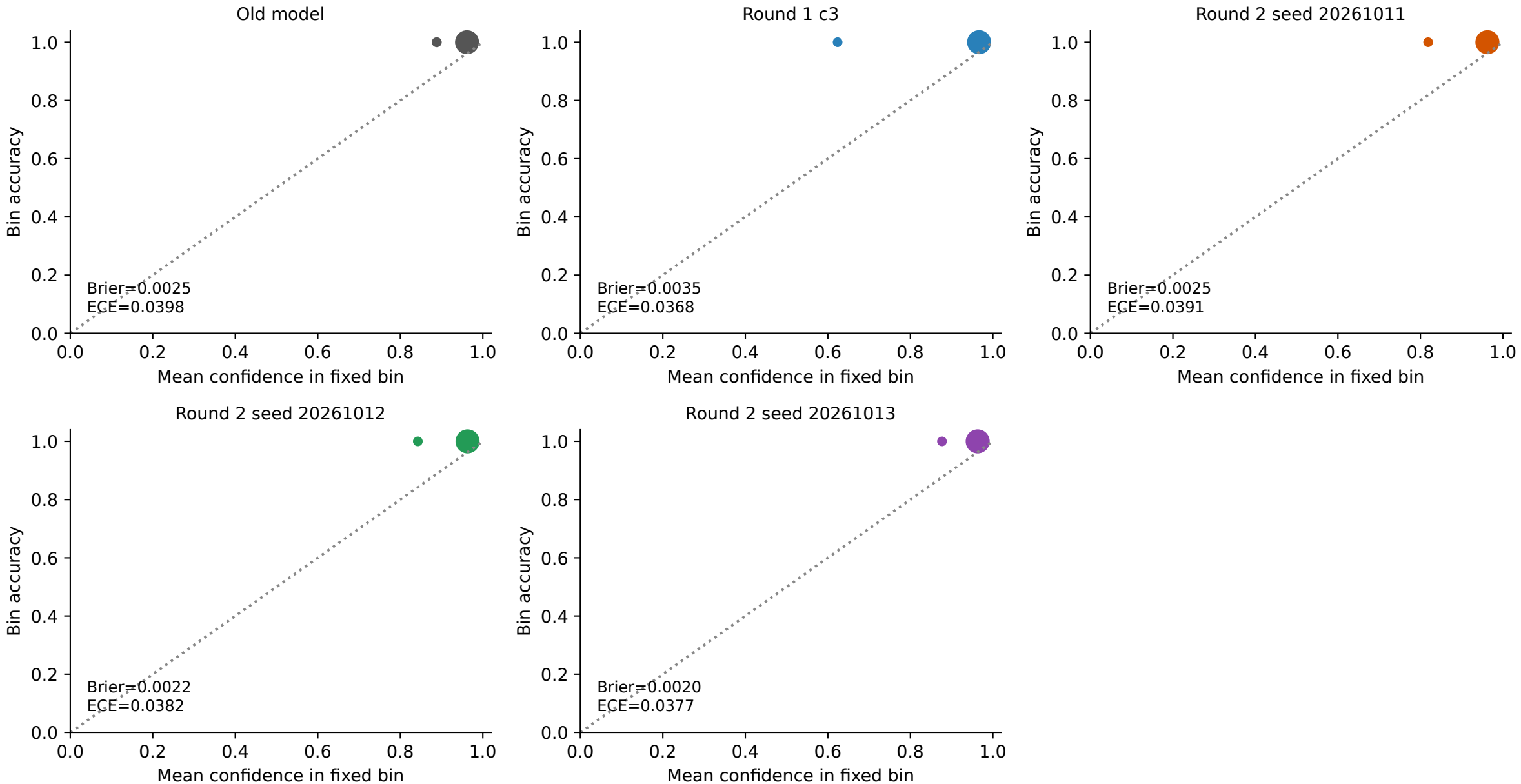

Reused reader benchmark (100) | confusion | n=100
Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Reader Kalapos
Reader Thamburaj
Reader Kanekar
Reference class
Predicted class
Mild
Moderate
Non
Very mild
Reference-row fraction
1.0
0.8
0.6
0.4
0.2
0.0
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Reused reader benchmark (100) | fixed stress label flip fraction
brightness
contrast
noise
blur
downsample
jpeg
label flip fraction (%)
Clean
0.9
1.1
0.01
0.02
0.5
192
160
90
70
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Same-device clean reference; all13 unchanged scenarios. Synthetic pixel stress is not MRI acquisition or clinical validation.

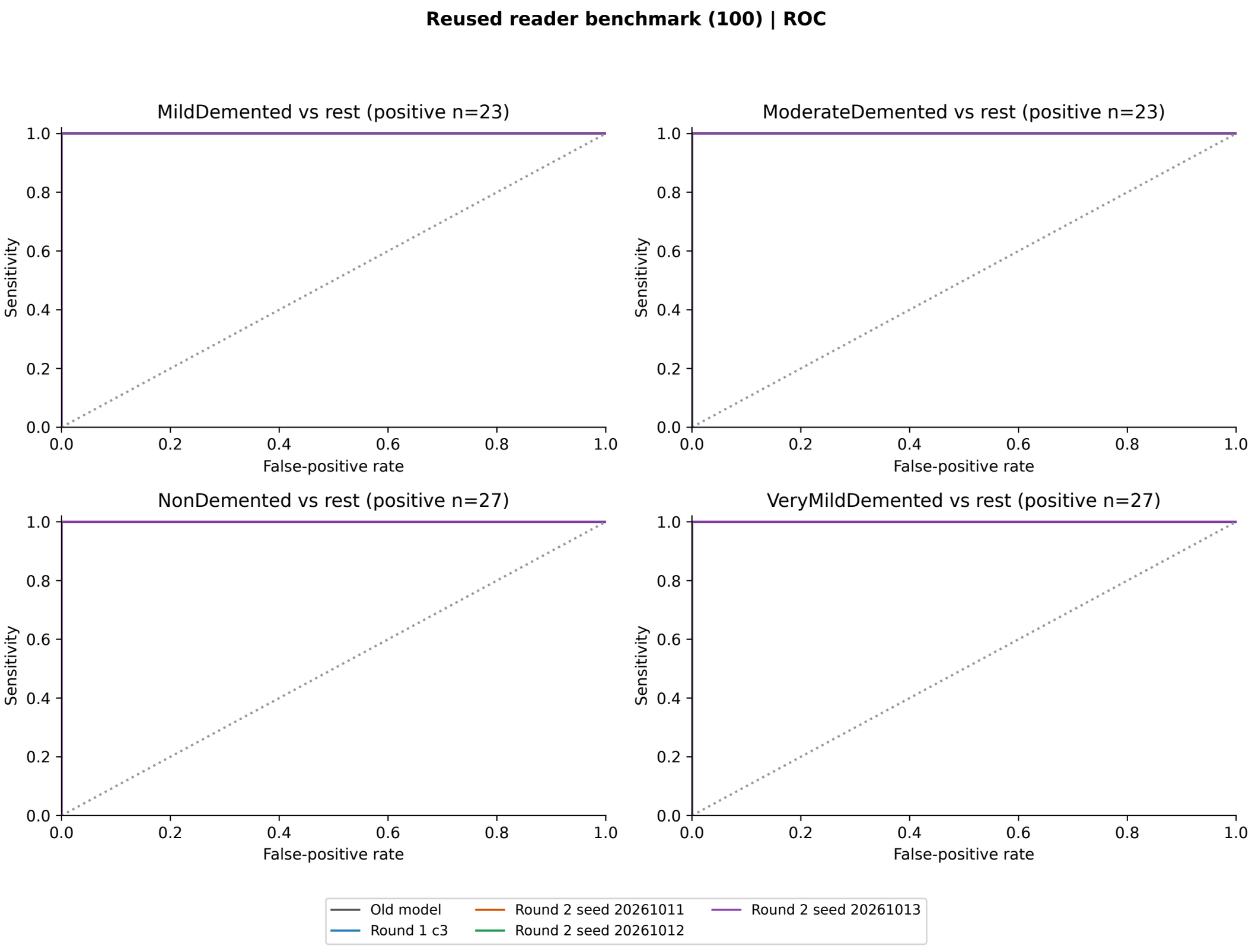
Reused reader benchmark (100) | Precision–recall



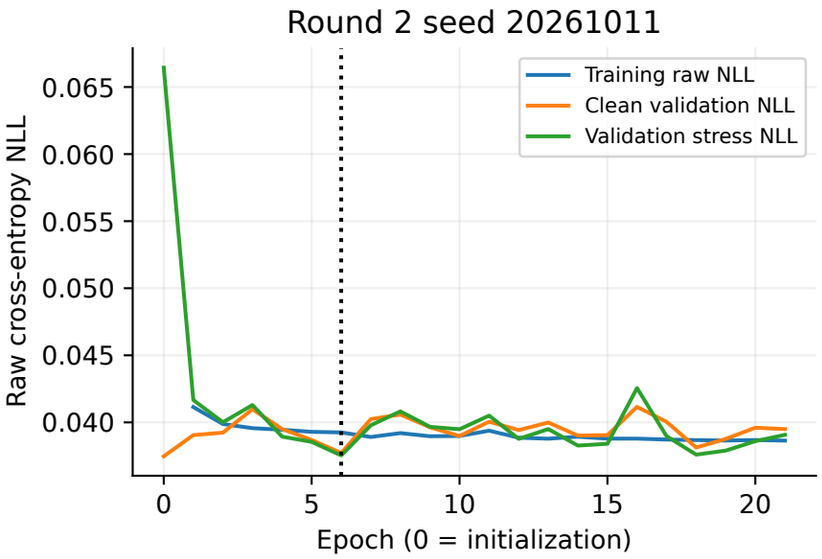
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

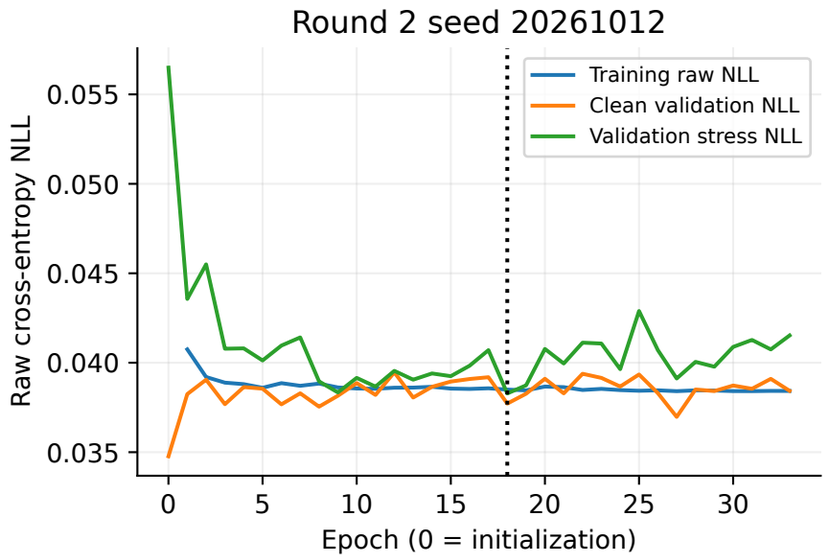


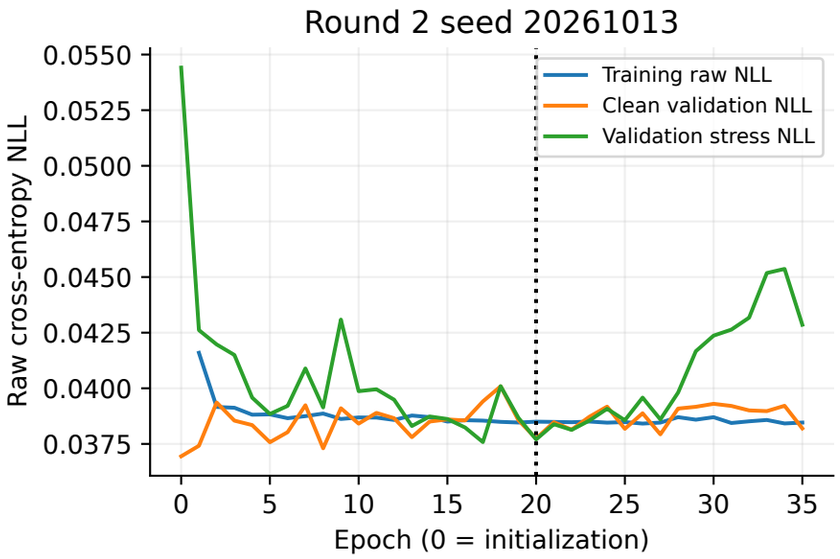
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

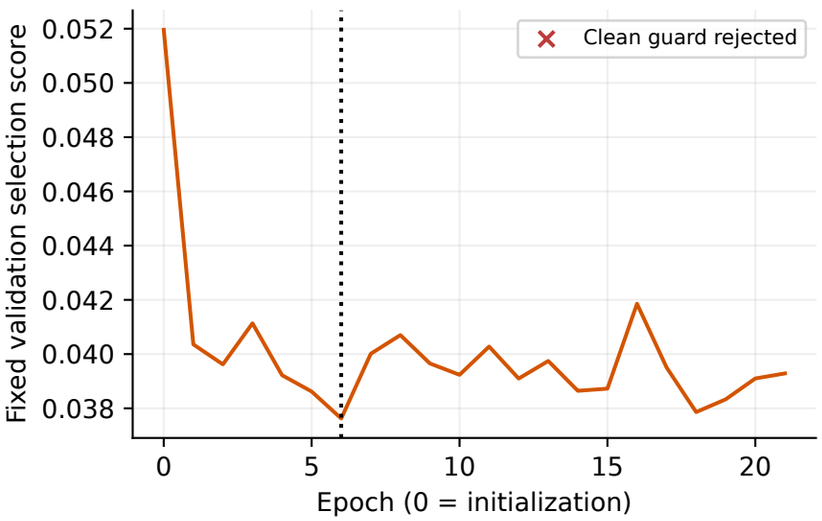

Reused reader benchmark (100) | ROC
MildDemented vs rest (positive n=23)
ModerateDemented vs rest (positive n=23)
NonDemented vs rest (positive n=27)
VeryMildDemented vs rest (positive n=27)
Sensitivity
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

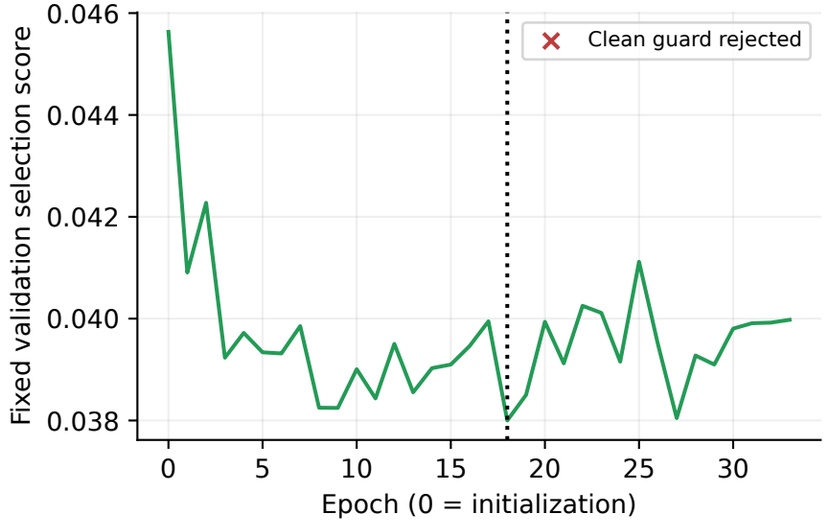
Round2 raw learning curves | all three runs, selected epoch marked

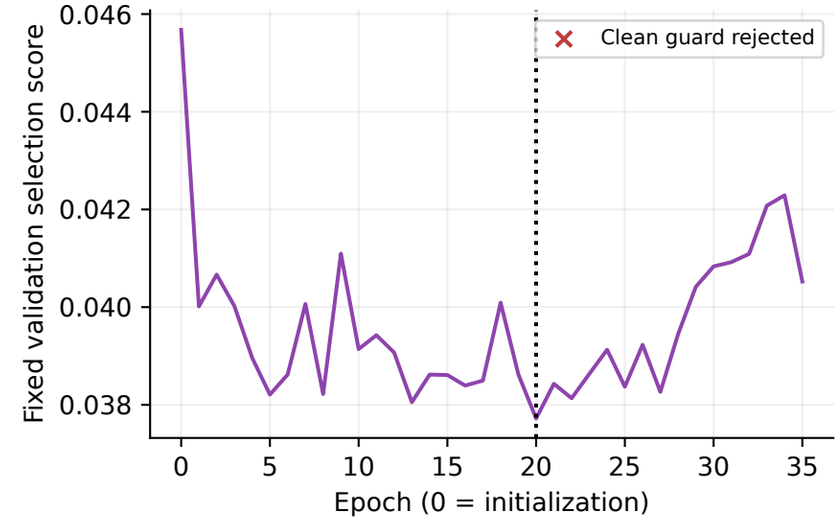


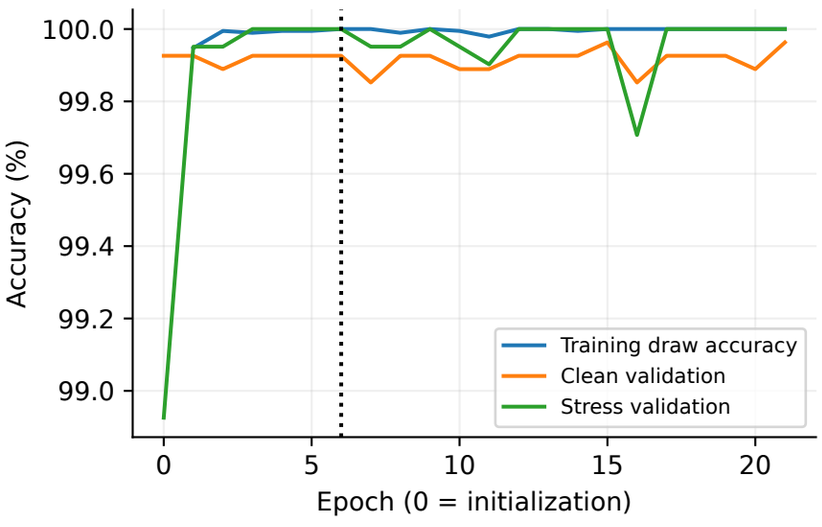


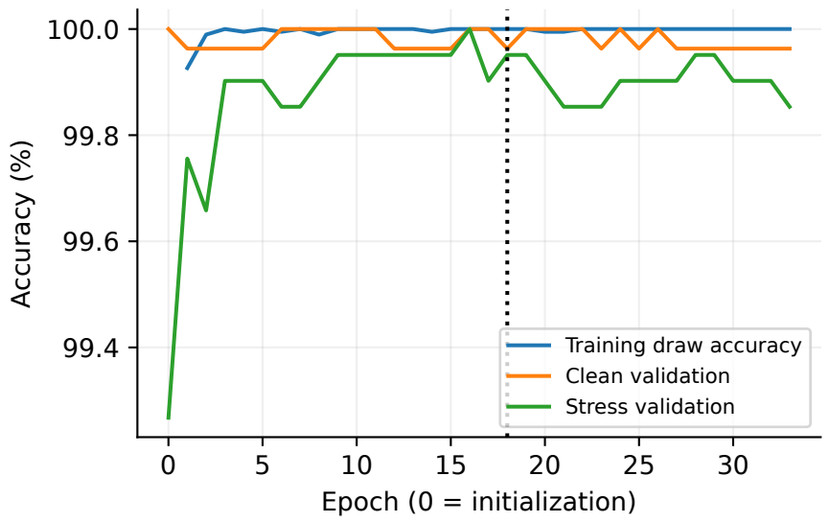


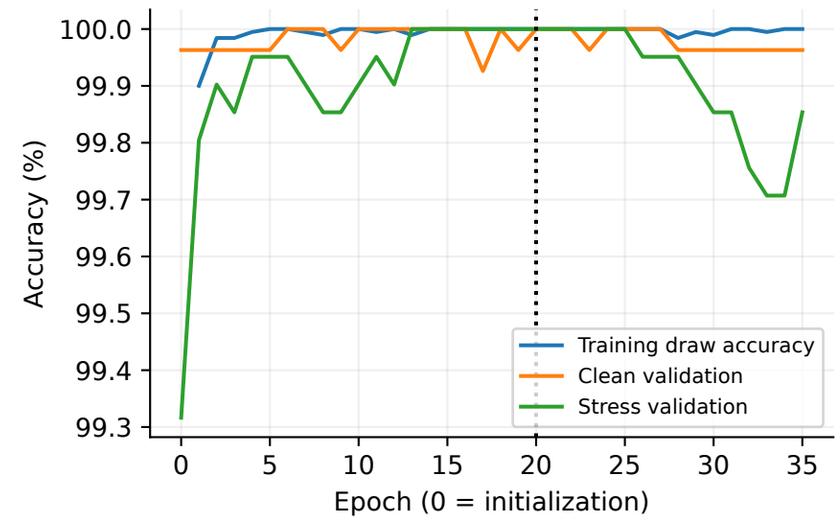




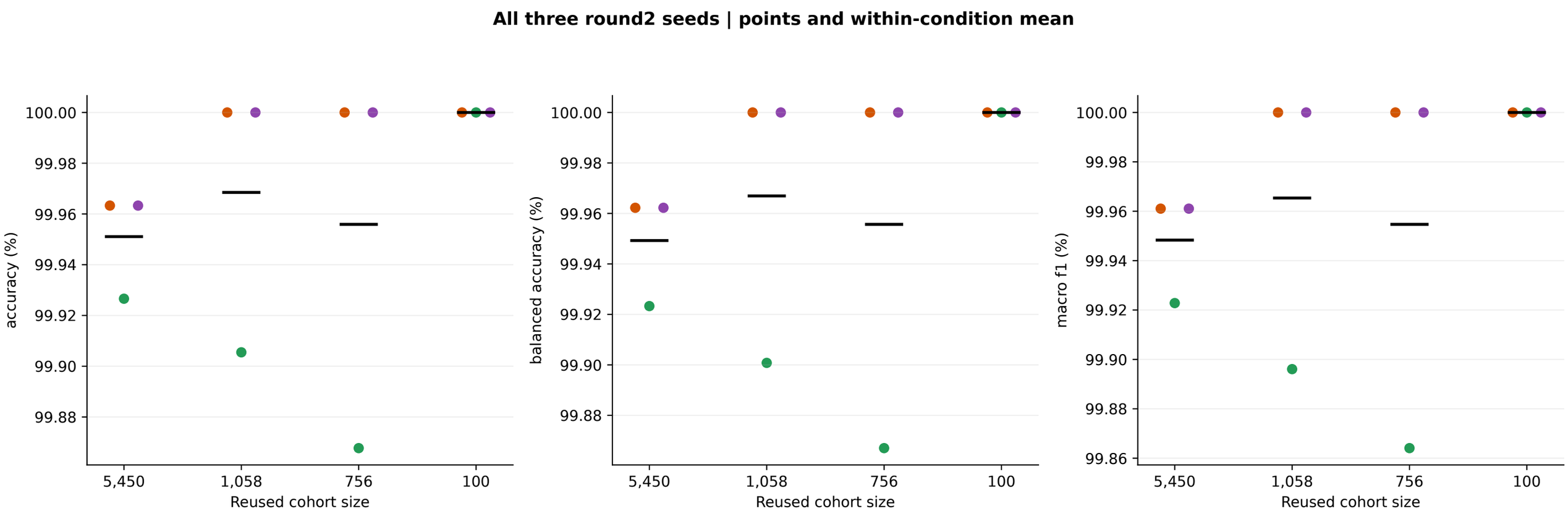




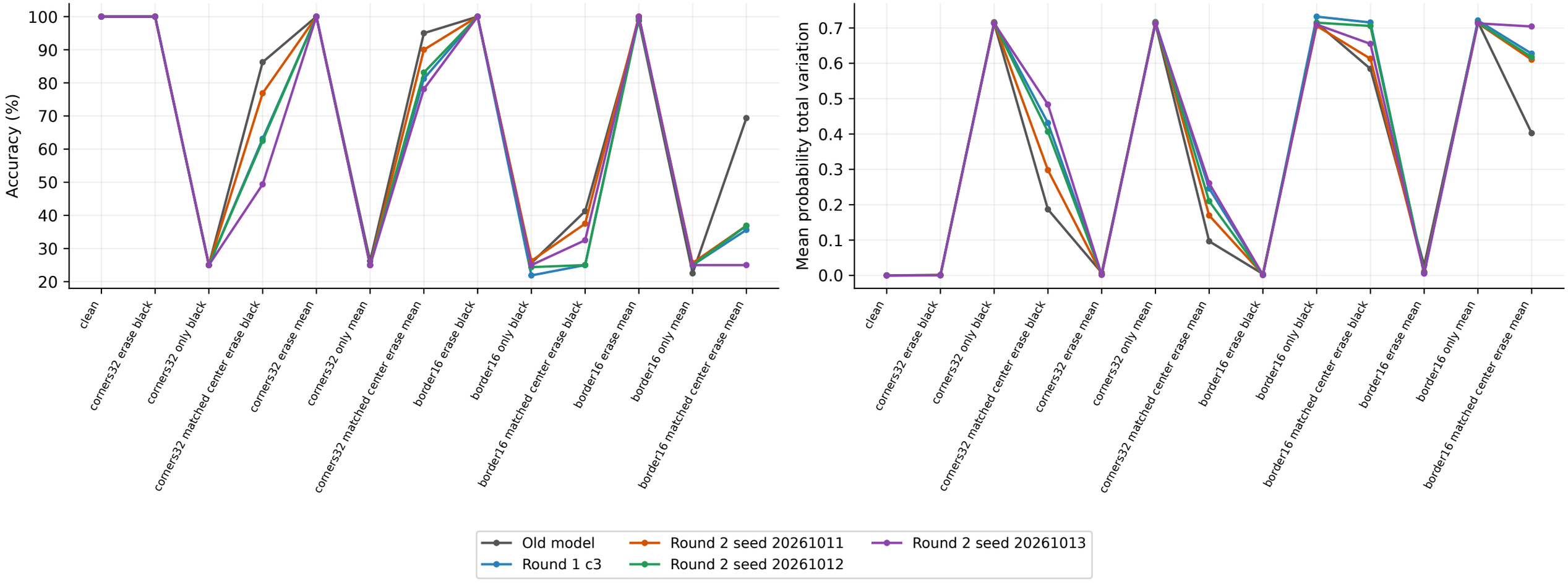




Training includes random augmentation; validation is fixed. Raw NLL differs from the weighted, label-smoothed optimization objective.
Epoch0 fallback, if selected, is retained; longer training is not evidence of a better model.



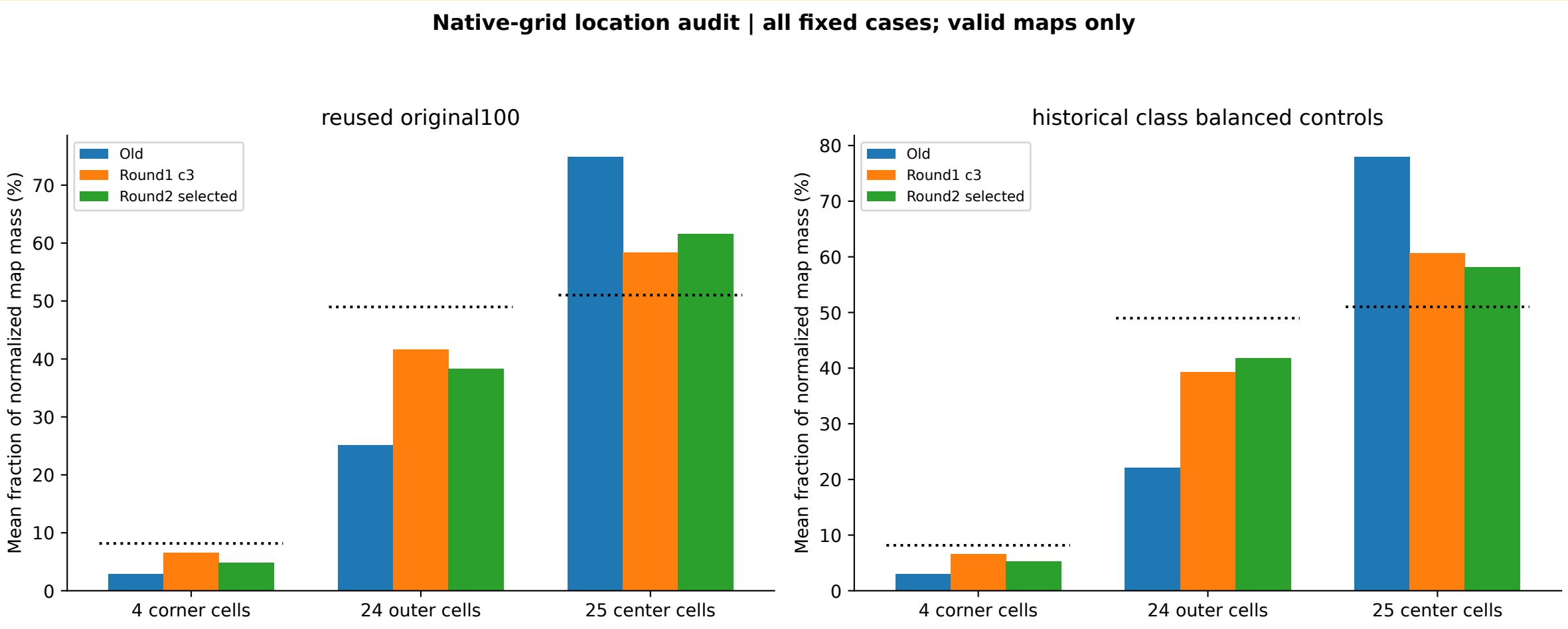
Seed variation is descriptive; the validation representative is fixed independently of these outcomes.



Erasure is out of distribution. Matched center areas and both fills are controls; these results cannot exclude all shortcuts.

DIAGNOSTIC ONLY — probability replay check failed (2/100; classes unchanged).

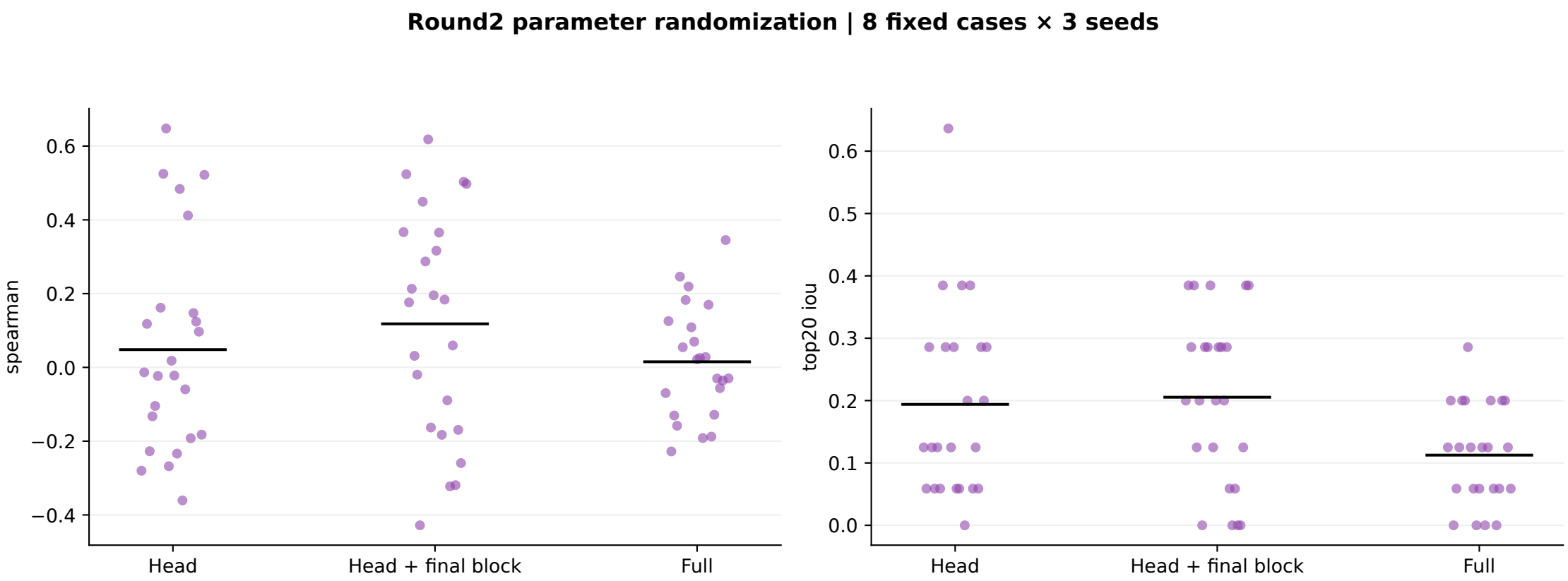
Tolerance 0.0001 unchanged. Not anatomical or clinical validation. See replay limitation.



Dotted lines: uniform area fractions. Outer cells overlap corner cells. Map mass is not anatomical probability or causal input importance.

DIAGNOSTIC ONLY — probability replay check failed (2/100; classes unchanged).

Tolerance 0.0001 unchanged. Not anatomical or clinical validation. See replay limitation.



Original predicted class stays fixed. Points exclude flat/failed maps, whose counts remain in randomization.csv. Model state is restored.